CALIFORNIA STATE UNIVERSITY
LONG BEACH
College of Natural Sciences
& Mathematics

Student Research Symposium



Book of Abstracts

Friday, September 19, 2025

Supported by: Jensen Student Access to Science and Math Center
& College of Natural Sciences and Mathematics

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California State University, Long Beach

Student Research Symposium



Funding support for this event provided in part by:
College of Natural Sciences and Mathematics, CSULB
Jensen SAS Center and G2 Lab

A Special Thanks to: Our Faculty Research Mentors, Jensen Student Access to Science and Mathematics Center, CNSM Academic Advising Center, Peer Mentors, The Pointe Conference Center, Campus Events, CNSM Tech Friendly, Daniel Ames, Lane Olsen-Cooper, Bianca Lopez Mendez, Brent Scheiwe, Brent Scheiwe, Lindsay Aymar, Dr. Deepali Bhandari (CNSM Associate Dean), Dr. Krzysztof Slowinski (CNSM Associate Dean), and Dr. Curtis Bennett (Richard D. Green Dean of College of Natural Sciences and Mathematics).

Without their support throughout the year, this event would not be possible.

Symposium Booklet and Event

The Student Research Symposium is held in The Pointe Conference Center Friday, September 19, 2025. This event, held by CSULB, College of Natural Sciences and Mathematics is open to undergraduate and graduate participation. The research being presented at this event is from on-campus research and/or from summer research experiences performed at other universities.

The symposium provides an opportunity for students to write abstracts, produce posters, and present research findings thereby bringing scientific and non-scientific communities together to share in ideas and discoveries. Students, staff, faculty, administrators, and community members attend this event and enrich the experience of all participants.

If this is your first time attending a symposium, feel free to walk around and ask the students questions about their research experience. We encourage any questions you may have about the research presented today. Thank you for attending our event.

The abstracts provided in this booklet are original works of students in our programs. Each abstract is included alphabetically by first author's first name.

Symposium Program

9:30am-9:50am:	Check-in and research & resource fair (until 1:00pm)
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Pizza will be served in the Summit Room at 11:30am and 12:30pm.

Project Abstracts

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1. **Electronic Collective Modes for Systems with Two Effective Masses**

Kaeden W. Russell and Andreas Bill, Ph.D.

Department of Physics & Astronomy, California State University, Long Beach, CA 90840

Collective electronic modes (plasmons) are essential to the screening of Coulomb interactions and the optical properties of metals. In conventional 3D metals these collective modes, called plasmons, are optical in nature and can be probed with light. There are other, typically low-dimensional systems where the plasmon energy varies linearly or as a square root with its wavevector q. David Pines proposed that in a 3D material with two species of charge carriers, one light and one heavy, acoustic plasmons may form. We present the derivation of the dielectric function for N bands in D dimensions. We apply this result to calculate the plasmon spectrum for two species of charge carriers and discuss the result and its implication for the observation of acoustic plasmons in 3D materials.

This work was funded through the CSULB Summer Student Research Assistantship, as well as through the Cal. State. Long Beach and the Ohio State University Partnership for Education and Research in Topological Materials, a National Science Foundation PREM, under Grant No. 2425133.

2. **Transfer Learning Analysis of Collective and Non-Collective Thomson Scattering Spectra**

Timothy Van Hoomissen (no longer presenting)

3. **Nonlinear current response contrasting high-harmonic response of trivial and nontrivial topological states**

Anthony Acebal, Sarah Grefe, Benedikt Fauseweh, Jian-Xin Zhu

This work theoretically studies the identification of nontrivial topology of topological semimetals in quantum materials. Current experimental tools, such as angle-resolved photoemission spectroscopy (ARPES) can be used to identify nontrivial topology, however, strongly correlated regimes require higher energy resolutions than even ultrahigh resolution ARPES probes can provide. We computationally study possible new diagnostic tools for topology using the nonlinear current response of a theoretical model to identify nontrivial topology in strongly correlated topological semimetals. Using intense field excitation, we calculate the nonperturbative nonlinear current response of the Weyl Kondo semimetal model (WKSM), which adequately models the topologically nontrivial Weyl-Kondo semimetal behavior of Ce3Bi4Pd3 and the topologically trivial Kondo insulator behavior of Ce3Bi4Pt3 through a single tuning parameter. We study the finite frequency response, contrasting the two states' nonlinear current response using a nonperturbative theoretical formalism which includes all orders of the impinging electric field. We analyze the calculated high harmonic generated current, and find differences between the WKSM and

Kondo insulator that may be differentiating signatures of topological states. This theoretical research aims to provide possible diagnostic tools that can be compared with experimental work on both the known Ce3Bi4Pd3 (WKSM) and Ce3Bi4Pt3 (Kondo insulator) and diagnose topological states of candidate materials.

This work was funded through the Cal. State. Long Beach and the Ohio State University Partnership for Education and Research in Topological Materials, a National Science Foundation PREM, under Grant No. 2425133

4. Effects of temperature on g-mode oscillations of neutron stars

David Morales-Zapien, Dr. Prashanth Jaikumar

We investigate finite temperature nuclear matter within the SU(2) chiral sigma model to construct an equation of state relevant for proto-neutron stars. By numerically solving the TOV equations, we obtained macroscopic observables such as mass and radius for various entropy/baryon ratios. We calculate the equilibrium and the adiabatic sound speeds, which encode information regarding the buoyancy of the matter inside the star. From this, we compute the g-mode oscillation spectrum of the star within the relativistic Cowling approximation. We find that the g-modes are suppressed as the entropy of the star increases, which has interesting implication for gravitational wave signals from neutron stars.

5. Thermodynamic Effects of B-L Non-conservation in Hot Quark Matter

Ethan Hsu[1], Thomas Klaehn[1]

[1]CSULB Department of Physics and Astronomy

Our universe is composed of charged particles. In addition to the classic electric charge, the charges describing these particles can include but are not limited to color, hypercharge, baryon number, and lepton number. Possible equilibration reactions are entirely defined by charge conservation among different particle species. This allows us to form equations governing the linear dependence between generalized chemical potentials of these "charges" and the individual particle chemical potentials. We can then use this to investigate thermodynamic fingerprints of baryogenesis in a model agnostic framework.

Acknowledgements: I would like to thank Prof. Klaehn for the opportunity to work on developing this formalism of his and providing feedback and aid every step of the way on my own research using it. I acknowledge the Gisela and Wilfried Eckhardt Endowment for Physics and Astronomy Summer Research Assistantship.

6. **Interacting Hadron Resonance Gas Model**

Tanner Massimino; Thomas Klähn, Ph.D.

The Hadron Resonance Gas (HRG) model has been widely successful in describing low energy, low density physics, such as that probed in particle colliders. In conventional HRG models, each species of meson and baryon is treated as a non-interacting ideal gas. However, at higher densities and temperatures, interactions between hadrons play an increasingly important role and must be considered explicitly. In this work, we extend the HRG model by incorporating density-dependent interactions mediated by scalar and vector boson exchange. These interactions are implemented through shifts in the effective masses and chemical potentials, with self-consistent gap equations and mean field approximations, allowing us to explore a larger area of the QCD phase diagram.

At high temperatures and low densities, the traditional domain of the HRG, the inclusion of interactions leads to faster growth of the pressure and trace anomaly. In the low temperature, high density regime, our model exhibits density driven mass melting and predicts a liquid–gas phase transition absent in conventional HRG models. While our model improves fit to lattice QCD data approaching the phase transition, it does not include explicit quark degrees of freedom and therefore cannot describe physics beyond this point.

Funding & Support: Thank you to the Keung Kai Luke Endowed Graduate Research Assistantship for supporting this research.

7. **Enhancing the Robustness for the Magneto-Optical Kerr Effect Magnetometry Set Up**

Brianna Christensen, Jiyeong Gu, Ph.D.

The Magneto Optical Kerr Effect (MOKE) magnetometry set up characterizes the magnetic switching behavior of magnetic thin films through the measurement of the modulation in the polarization of reflected light. Recently, our MOKE magnetometry set up has been upgraded to introduce 4f alignment and accommodates computer controlled fast-steering mirror to achieve scanning capability. Adding 4f alignment for the incident laser beam cleans and focuses the incident beam on the sample surface and filters out the noise to improve the quality of the beam. The MOKE hysteresis loops are measured using the LabView program. We discovered two prominent issues related to magnetic field control in the program. Firstly, an overly saturated magnetic field. And secondly, a linear relationship of the coercivity with the ramping rate. We tried resolving this issue by carefully checking out the effect of each parameter to the resulting hysteresis loop measurements. During this process, we measured the 20 nm SmCo and 200 nm SmCo nanocap layer deposited on 890 nm nanosphere templates. These MOKE hysteresis loops were compared with ones measured using Alternating Gradient Magnetometer.

This work was funded through the Cal. State. Long Beach and the Ohio State University Partnership for Education and Research in Topological Materials, a National Science Foundation PREM, under Grant No. 2425133.

8. Building a Magneto-optical Kerr Microscope by Modifying a Polarizing Microscope

Andy Le Blanc, Jiyeong Gu

The magneto-optical Kerr effect (MOKE) magnetometry method using a laser is an effective characterization tool capable of measuring local magnetic hysteresis loops of magnetic samples in sub-micron size. Originally, we tried to upgrade our current MOKE hysteresis loop measurement set up to a prototype Kerr microscope that can also capture magnetic domain images. However, it turned out that it is difficult to optimize a laser system for both purposes of measuring intensity for a MOKE hysteresis loop and imaging contrasts of magnetic domains. Therefore, we developed a dedicated MOKE microscope by modifying a regular polarizing microscope for the purpose of imaging magnetic domains of patterned magnetic thin film structures, nanometer/sub-micron-size samples. Permalloy (Py) and SmCo nanocaps were fabricated on nano/microsphere templates using a pioneering fabrication method of nanosphere lithography. By using the MOKE method on our samples, we observed contrasts that reflected the presence of magnetic domains of the micrometer scale. By measuring both hysteresis loops and magnetic domains we expect to provide an understanding of the magnetic characteristics of three-dimensional magnetic nanostructures such as our nanocap thin films.

We acknowledge support from the APS Bridge Program and this work was funded through the Cal. State. Long Beach and the Ohio State University Partnership for Education and Research in Topological Materials, a National Science Foundation PREM, under Grant No. 2425133.

9. Hysteresis and FORC Characterization of SmCo and Permalloy Nanocap Thin Films

Alan Mejia, Dr. Jiyeong Gu

Recently, interest in complex 3D nanomagnetic systems has increased due to new opportunities for improved performance in device applications such as data storage devices, spintronics, and biological or chemical sensors. Nanomagnetic systems with complex geometry also exhibit novel magnetic behavior induced by curvature. A nanocap thin film, which is a thin magnetic layer deposited onto spherical silica templates, is a good example of a 3D nanomagnet. 20 - 30 nm of Permalloy (Py) and Samarium Cobalt (SmCo) nanocaps were fabricated on top of hexagonally close-packed single-layer nanosphere templates. The nanosphere template was made by the dry rubbing method. Magnetic characterization was performed using hysteresis loops and First Order Reversal Curve (FORC) diagrams. Hysteresis loops revealed differences in coercivity and switching behavior, while FORC diagrams helped separate magnetic switching mechanisms,

interactions, and coercivity distributions. Comparing hysteresis loops between samples revealed that increasing Py thickness from 20 nm to 30 nm on uniform silica spheres resulted in higher coercivity (12 Oe vs. 27 Oe) and more gradual switching. Comparing 30 nm Py on curved silica to flat silicon showed that curvature significantly affects the coercivity (27 Oe vs. 3 Oe) and overall magnetic switching behavior.

This work was supported by the Partnership for Research and Education in Materials (PREM) program between California State University, Long Beach and The Ohio State through a grant from the National Science Foundation under Grant No. 2425133. We gratefully acknowledge the support from the Gisela and Wilfried Eckhardt Endowment for Physics and Astronomy Summer Research Assistantship.

10. Fabrication of micropillar Josephson junction with Niobium Nitride and Permalloy/Samarium Cobalt exchange spring magnet using Microwriter ML3

Adam Bowen, Oscar Padilla, Jiyeong Gu

The superconducting proximity effect in the superconductor(S)/non-superconductor hybrid system has been studied extensively. Josephson junctions are devices where non-superconducting material is sandwiched between the two S layers and the interaction between the two S layers is measured by critical current. We fabricated micropillar exchange spring coupled Josephson junctions (μ-ESCJJs) with a ferromagnetic (F) layer in the middle and attempted to observe the odd triplet effect in a S/F/S system. For the vertical transport measurement, smaller lateral sizes of a junction (of a micrometer or less) are required. Earlier efforts on junction fabrication faced a few complications, for example, low superconducting critical temperature and difficulty in defining lateral dimension of the junction due to the irregular distribution of nanosphere mask. Originally, Niobium (Nb) was used as the S layer of the junction, until it was decided that the critical temperature was too low. The change from Niobium to Niobium Nitride (NbN) sees an increase in critical temperature, thus leading to a higher temperature range of the superconducting state for the Josephson junction. Changes made in lithographic practices using Microwriter ML3 have allowed for more precise electrode and micropillar formulation. The result of the cleaner lithographic process should allow for improved devices suitable for transport measurement.

This work was funded through the Cal. State. Long Beach and the Ohio State University Partnership for Education and Research in Topological Materials, a National Science Foundation PREM, under Grant No. 2425133.

11. Controlled Dilution of Copper Spins in Phthalocyanine Thin Films for Electron Paramagnetic Resonance

Alexander Goytia Fajardo, Emmy Freudenrich, Rob Claasen, Ezekiel Johnston Halperin, Thomas Gredig

Department of Physics and Astronomy, California State University Long Beach, CA 90840, USA Department of Physics, The Ohio State University, Columbus, OH 43210, USA

Electron paramagnetic resonance (EPR) provides a sensitive probe of hyperfine interactions in molecular solids. In copper phthalocyanine (CuPc), these interactions originate from localized copper spins, but concentrated spin systems often obscure fine details. To overcome this limitation, we demonstrate approaches for preparing diluted CuPc thin films by co-depositing metal-free phthalocyanine (H_2Pc) as a non-magnetic host matrix. Two complementary fabrication methods were employed to control the concentration of paramagnetic copper centers. The resulting films were characterized using x-ray diffraction (XRD) to assess crystallinity and atomic force microscopy to evaluate surface morphology. The large relative XRD peak ratios are consistent with a large concentration of H_2Pc. We also discuss common challenges in employing diluted thin-film systems for high-resolution EPR investigations of hyperfine interactions in molecular semiconductors.

This work was funded through the California State Long Beach and the Ohio State University Partnership for Education and Research in Topological Materials, a National Science Foundation PREM, under Grant No. 2425133. One of the authors (A.G.F.) would like to acknowledge partial support from the Google Summer Research Assistantship.

12. Creation and Control of Crystals of Topological Knot Solitons

Holland Karaghiaulleian[1,2], Gewei Chen[2], Ivan I. Smalyukh[2]

[1]Department of Physics and Astronomy, California State University Long Beach, [2]Department of Physics, University of Colorado, Boulder

The beginnings of knot theory can be traced back to Lord Kelvin, who proposed the vortex-knot structure of atoms in a background ether. Today, knot theory not only remains to be an important branch of pure mathematics, but extends its applications across many fields of physics, including hydrodynamics, superconductors, Bose-Einstein condensates, and liquid crystals. However, experimental observations and manipulations of these knotted structures is challenging because of their transient nature. Here, we realize a controllable and stable crystal system of topological solitons, called ``heliknotons,'' embedded in a chiral liquid crystal background. A single heliknoton is created by locally melting the chiral liquid crystal medium into the isotropic phase then allowing it to relax to a stable knotted soliton in a weak external electric field. Arrangements of heliknotons can then be formed via holographic laser tweezers capable of manipulating heliknotons. We probe the morphology, evolution, and dynamics of single heliknotons and of heliknoton lattices by tuning

parameters of the host medium, such as the helical pitch, the depth and boundary conditions of the cell, and the applied electric field strength. We present a wide variety of behavior prompted by small perturbations in the system parameters, showcasing the high tunability of heliknoton systems and different avenues for controlling their self-assembly.

Funded by NSF Physics/JILA REU at University of Colorado, Boulder

13. Fabrication of Graphene and hexagonal Boron Nitride

Narith Chan, Said Ortega, Eric Corona-Oceguera, Claudia Ojeda-Aristizabal

Graphene and hexagonal boron nitride (h-BN) are two-dimensional materials with unique structural and electronic properties. We exfoliated graphene and h-BN monolayers on silicon wafers using a tape transfer method. The wafers were cleaned with compressed air and heated to 180°C to 220°C before exfoliation. We have a tape with thin flakes of graphite, onto which we apply the wafers, after a short while at a high temperature. Due to its thinness, monolayer graphene looks nearly transparent and requires careful inspection of a wafer through the microscope. Once a flake of decent size is found, they are further verified through Raman Spectroscopy, which reveals the characteristic graphene peak. This approach allows the successful isolation of monolayer graphene, which will then be stacked onto h-BN to create a device because of its unique heterostructure properties.

Acknowledgements: The primary funding for this work was provided by the U.S. Department of Energy, Office of Science, Office of Basic Energy Sciences under contract DE-SC0018154 for sample fabrication. Raman characterization was funded by the California State University, Long Beach and the Ohio State University Partnership for Education and Research in Topological Materials, and the National Science Foundation PREM, under Grant No. 2425133

14. Proximity-Induced Electronic and Magnetic Effects in Phthalocyanine/Graphene/h-BN Heterostructures

Eric Corona-Oceguera, Narith Chan, Claudia Ojeda-Aristizabal

Graphene is an ideal building block of complex heterostructures. Normally, in proximity to molecular thin films its mobility is compromised. However, our recent results show that graphene in proximity to a molecular thin film of Copper Phthalocyanine (CuPc) preserves its mobility while transferring an important amount of electrons to CuPc (~10^12/cm^2). Additionally, we have found that weak localization in bilayer graphene is restored by the presence of the CuPc molecules.

Our heterostructures are fabricated by transferring graphene on hexagonal boron nitride (h-BN) and depositing through thermal evaporation of a phthalocyanine thin film. We plan to perform experiments using Manganese Phthalocyanine (MnPc), a similar molecule that has a higher magnetic moment than CuPc and an unpair electron of S=3/2. MnPc is one of the

few Pcs to develop long range magnetic order at low temperatures. We expect to see in electronic transport experiments on a MnPc / graphene / h-BN heterostructure signature of the proximity effects with a ferromagnetic molecular thin film, measured in the quantum coherence regime.

Funding: The primary funding for this work was provided by the U.S. Department of Energy, Office of Science, Office of Basic Energy Sciences under contract DE-SC0018154 for electronic transport measurements, data analysis, sample fabrication and Raman characterization. The deposition of the molecular thin film was funded by the Cal. State. Long Beach and the Ohio State University Partnership for Education and Research in Topological Materials, a National Science Foundation PREM, under Grant No. 2425133. 4D-STEM measurements were possible thanks to BioPACIFIC, an NSF Materials Innovation Platform (DMR-1933487).

15. The Fabrication of a $PtTe_2$ and $RuCl_3$ Van-deer-Waals Heterostructures for Electronic Transport

Joshua Luna, Michelle Mckenzie, Dr. Claudia Ojeda-Aristizabal

PtTe2 is an air-stable exfoliatable ferromagnet that features a type-II Dirac cone. Electrons follow a linear dispersion relation that is different along one of the momentum directions. The type-II Dirac Cone of $PtTe_2$ is .8 eV below the Fermi energy, which makes it hard to access through electronic transport experiments. $RuCl_3$ is a material that is known to be a very good electron acceptor. We propose a $PtTe_2$ and $RuCl_3$ heterostructure to hole dope $PtTe_2$ in order to potentially observe signatures of type-II fermions under electronic transport. We discuss the methods involved in creating this heterostructure and isolating thin flakes of $PtTe_2$ and $RuCl_3$. Which contributes to the goal of overall project.

Sample Fabrication was supported by the US Department of Energy, Office of Science, Office of Basic Energy Sciences under award number DE-SC0018154.

Sample characterization through Raman spectroscopy was funded through the Cal. State. Long Beach and the Ohio State University Partnership for Education and Research in Topological Materials, a National Science Foundation PREM, under Grant No. 2425133.

16. Fabrication and Electronic Transport Measurements of Thin Layers of the Type-II Dirac Semimetals Platinum Ditelluride

Michelle McKenzie[1], Eric Corona-Oceguera[1], Movindu Dissanayake[1], Chenchu Yakasiri Saravanan[1], Archibald J. Williams[2], Joshua Goldberger[2], and Claudia Ojeda-Aristizabal[1]

[1]Department of Physics and Astronomy, California State University, Long Beach,
[2]Department of Chemistry and Biochemistry, The Ohio State University

Two-dimensional transition metal dichalcogenides (TMDs) such as $PtTe_2$ have recently gained attention for their unique structural, optical, and electronic properties. In this

research, thin $PtTe_2$ electronic devices were fabricated through mechanical exfoliation and characterized using several techniques including energy dispersive spectroscopy (EDS), atomic force microscopy (AFM), and Raman spectroscopy. Exfoliation with 1007R adhesive tape combined with substrate heating provided the highest yield of device-quality flakes. EDS confirmed the Pt:Te stoichiometry, while AFM established correlations between flake thickness and optical contrast. Raman spectroscopy identified the characteristic *E*g (~110 cm^{-1}) and *A*1g (~157 cm^{-1}) vibrational modes, with thickness-dependent spectral shifts consistent with literature. Together, these methods establish reliable approaches for identifying and characterizing thin $PtTe_2$ crystals for device fabrication and future electronic transport measurements.

This project is supported by PREM, The U.S. Department of Energy and the APS Physics Bridge Program.

17. Characterization of atomically thin PtTe2 and their Cr alloys for electronic transport devices

Chenchu Yakasiri, Joshua Luna, Movindu Dissanayake, Stefan Lucero, Anise Mansour, Michelle Mckenzie, Archibald J. Williams, Joshua Goldberger and Claudia Ojeda-Aristizabal

We show our currents results characterizing the thickness of thin crystals of PtTe2 through atomic force microscopy (AFM) and Raman spectroscopy. Additionally, we present our current efforts at fabricating electronic devices of the CrxPt1-xTe2 family of materials. By informing our sample fabrication with these two techniques, we are able to improve our sample throughput for electronic transport measurements at low temperatures.

This work was supported by the US Department of Energy, Office of Science, Office of Basic Energy Sciences under award number DE-SC0018154. Funding for travel was provided by the Cal. State. Long Beach and Ohio State University Partnership for Education and Research in Hard and Soft Materials, a National Science Foundation PREM, under Grant No. 2425133.

18. Defect Tolerance in Interlocked Ring Polymer Sheets: How Solvent Quality Modulates Structural Response to Ring Deletion

Jason Spada, Jacob Breese, Lisa Hall

Rings of DNA can be interlocked to create chainmail-like sheets, such as those uniquely found in certain organisms called kinetoplasts. Prior studies have explored the topological behavior of these sheets, showing that the type of ring linkage changes the sheet's preferred shape in solution, with some initial work on how their structure changes with solvent conditions. Here, we investigate how the structural features of the sheets, such as average size and curvature, change as a function of ring removal, and examine how this changes for incomplete sheets (with rings deleted randomly from an initially complete lattice of interlocked rings). Specifically, using coarse-grained simulations that treat the rings as

bead-spring chains, we analyze the structure's radius of gyration and shape in implicit solvent of various effective solvent qualities.

Acknowledgement: This work was supported by the Center for Emergent Materials (CEM), Partnerships for Research and Education in Materials (PREM), and the National Science Foundation (NSF). We gratefully acknowledge the Ohio Supercomputer Center (OSC) for providing high-performance computing resources. The author expresses deep gratitude to Dr. Lisa Hall and Jacob Breese for their continuous guidance and support throughout this research project.

19. Alpha-Abundance Bimodality in the APO-K2 Catalogue and NIHAO Milky Way--Mass Galaxies

Ivy Boiko, Joel C. Zinn, Lucy Lu, Jack Warfield

Since its discovery more than two decades ago, the physical origin behind the formation of two seemingly distinct stellar populations of 1) high-alpha stars — which often have more than twice the amount of alpha elements (e.g., O, Mg, Ca) than the Sun — and 2) low-alpha stars has been contested. Several models for the origin of these two stellar populations have been proposed, including variable star formation rates, merger events, and/or stellar migration. By examining the intersection of stellar chemical abundances, ages, and positions in the Galaxy, it becomes possible to make stringent tests of these models. Here we perform a preliminary comparison between observed stellar data and a merger-dominated galaxy evolution model, with a particular focus on the metallicity distribution function (MDF) of the high-alpha population of red giant branch stars. It has been previously shown that the high-alpha population is uniformly old and has a spatially constant MDF. Upon analysis of the observational data from APO-K2, we observe an unexpected variation of the metallicity distribution with radius in this high-alpha population. The observed radial dependence of the MDF persists after examining potential correlations with age and selection biases in the targeting of the APO-K2 data. Our comparison to models shows that these trends are not present in a representative galaxy from the NIHAO-UHD cosmological project. This continuing work motivates further exploration of potential selection functions in the APO-K2 dataset, as well as future comparisons against the broader suite of Milky Way–mass galaxies from NIHAO. We also aim to extend our analysis to include stellar kinematics, which would provide the first comparison of NIHAO predictions with data in combined chemical abundance, age, and kinematic space.

Acknowledgements: This work was supported by the Google Summer Research Assistantship, awarded by the Physics and Astronomy Department at California State University, Long Beach.

20. Toward A Pixel-Level Saturated Photometry Pipeline for Asteroseismology with the Nancy Grace Roman Space Telescope

Lauryn James

The Nancy Grace Roman Space Telescope promises to usher in a new era of asteroseismology, which relies on precise, months-long measurements of stellar brightness, by providing the first time-resolved view of the center of the Galaxy from a space-based infrared survey. Roman will do so through its Galactic Bulge Time Domain Survey, which has been recently shown to be able to detect asteroseismic parameters for hundreds of thousands of red giant stars (Weiss et al. 2025). This previous work demonstrated that approximately 20% of the detections will be in the saturated regime of Roman's detectors, which means the brightness measurements will be too inaccurate to perform asteroseismic measurements. Here, we present an approach to identify and remove saturated pixels from Roman's raw photometric data from its Wide Field Instrument, which is a promising way forward to recover accurate light curves in the saturated limit. We implement this approach in a pixel-level photometric pipeline applied to simulated photometry from single stars using the Roman team's Roman I-Sim software. We also implement in Roman I-Sim an updated treatment of persistence, which results in lingering flux in pixels from previous observations, and which can reduce the precision of photometry. Future work will include generalizing this pixel-level photometry framework for application beyond single stars and to entire synthetic stellar populations in the bulge. The resulting saturated photometry pipeline promises to not only support robust asteroseismic analysis but also to strengthen Roman's broader mission goals in time-domain astrophysics and exoplanet detection.

This work was supported under NASA award 80NSSC24K0091. The Roman Asteroseismology Wide Field Science Team is composed of Tim Bedding, Kaili Cao, Noah Downing, Scott Gaudi, Marc Hon, Daniel Huber, Jennifer Johnson, Matthew Penny, Marc Pinsonneault, Claudia Reyes, Sanjib Sharma, Dennis Stello, Trevor Weiss, Lauryn James, and Joel C. Zinn. We thank Tom Barclay, Joshua Schlieder, and the Roman Team for their insight on the project.

21. Identifying Significant Photometric Periodicity in Rotating Kepler Giants

Rachel M. Perelgut, Dr. Joel C. Zinn

After decades of study, the magnetic field generation mechanism (dynamo) of main sequence (MS) stars like the Sun remains contested. An important observable in testing these models is the years-long variability of magnetic field strength. In particular, stellar dynamo theories make testable predictions of the relationship between periodic photometric variability caused by magnetic cycles and the stellar rotation rate. While there exists some indication for magnetic cycles in red giant branch stars, it is not clear if their cycles have the same dependence on stellar rotation as do their less evolved main sequence (MS) star counterparts. Here, we search for magnetic variability in a sample of red

giants using photometric data from NASA's Kepler mission. Using an open source prewhitening software, we search for periodicity on year-long timescales, and report the discovery of over a dozen stars with significant periodic photometric variability. We discuss our results' implications for a connection between the dynamo mechanism in MS and red giants, including a potential analog to the inactive magnetic cycle—rotation rate branch of the MS.

Funding acknowledgement: This project was supported in part by a CNSM Faculty Small Grant

22. Understanding Monsoon Dynamics in East Africa (Madagascar) and the Indian Ocean.

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Current climate data depicts a small fraction of total climate records on Earth. Furthermore, there is a lack of observational data for long term monsoon dynamics in the Southern Indian Ocean. Paleoclimatology reveals the past of Earth's climate before instrumentation existed to record changes in Earth's systems often on millennial time scales. Speleothems, such as stalagmites, help researchers and scientists identify, analyze, and record the changes in oxygen isotope values, which can tell us about precipitation patterns and changes in sea-surface temperatures on Earth. This project aims to analyze spatial climate data using Python and to compare paleoclimate records to climate models, so we can better predict future climates. Our team conducted three major comparative analyses when visualizing our climate data: (1) the TraCE-21ka model with Global Precipitation Climate Project (GPCP) observations, (2) the TraCE-21ka and iTraCE-21ka models with each other, and (3) both TraCE and iTraCE simulations with speleothem records from Madagascar. By visualizing spatial climate data in Python, our team identified significant discrepancies in seasonal precipitation over the Southern Indian Ocean during the Last Glacial Maximum period and Heinrich-Stadial 1 period. While the TraCE simulations reproduce proxy signals during the Henrich-Stadial 1 period, the iTraCE simulations show inconsistencies with both the AB-12 and ABC-1 speleothem records. To further improve the accuracy of climate models, we should isolate the iTraCE CCSM3 single-forcing simulations and compare them with the TraCE CCSM3 single-forcing simulations and speleothem records. Moreover, collecting more paleoclimate archives will provide more robust constraints for model validation and improvement.

This research project was supported by the Massachusetts Institute of Technology (MIT) Office of Graduate Education and the MIT Summer Research Program (MSRP).

23. How Can Science Help Us Answer Archaeological Questions About Lechaion, Ancient Corinth?

Nathan Chan[1], Aiyana Martel[1], Christine Solis[1], Samuel Oliver[2], Darryl Acosta[3], Marrissa Martinez, Tau Miller, Ismael Sandoval, Lora Stevens, Ph.D.[1], AJ White, Ph.D.[1]

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Lechaion Harbor was the major port for the ancient city of Corinth, Greece. It existed for a millennium serving the Greeks and was later re-established as a Roman colony. Little is known about the types and intensity of human activities at the port. Using a suite of geochemical analyses, our project seeks to use scientific techniques and relate them to known events to determine how the ancient Greeks and Romans used and impacted the landscape around this harbor. Climate can have a profound effect on human activity and thus provides important context for human activity. Using stable isotopic ratios of shells collected from the site's sediment, past changes in climate can be inferred. In addition to changes in climate, the residents of Lechaion experienced a number of natural (ex. Earthquakes) and human (ex. war) disasters. The population is expected to have decreased after such events, but to what extent is unknown. The analysis of fecal biomarkers, such as stanols and bile acids, has been used to determine relative changes in human populations over time. By relative changes in fecal biomarkers with known disruptions, we will determine how these events impacted the number of people using and living at the port. These markers can also provide evidence for agriculture and animal husbandry. Furthermore, natural catastrophes like forest fires and war can be identified through high concentrations of charcoal, which also provide radiocarbon dates. It is expected that such catastrophes would leave higher concentrations of charcoal on the landscape, which can also be supplemented by an analysis of what burned. Finally, it is suspected that smelting may also have occurred in Lechaion. Heavy metal analysis can track the intensity of these activities and possible changes in what ores were smelted. These analyses require a measure of total carbon provided by elemental analysis. In summary, all of these individual pieces can be brought

together to create a picture of human activity in ancient times where more conventional records do not exist.

This project is supported by the National Science Foundation and the Lechaion Harbor and Settlement Land Project.

24. At The Periphery of Lechaion, Greece: Preliminary Findings from the 2025 Field Season

Samuel Oliver, Ismael Sandoval, Daryll Acosta, Aiyana Martel, Christine Solis, Marissa Martinez, Tau Miller, Lora Stevens, Ph.D, AJ White, Ph.D.

Lechaion Harbor was the major port for the ancient city of Corinth, Greece. It existed for a millennium serving the Greeks and later was re-established as a Roman colony. The site is currently being excavated by the Lechaion Harbor and Settlement Project, a collaboration between the Greek Antiquities Ministry and CSU Long Beach Classics program. Our project investigates a heretofore unstudied part of Lechaion to learn more about past land use and environmental conditions in the area. One of the Subsurface images from a geophysical methods used was ground penetrating radar (GPR), a geophysical technique that uses radio waves to image subsurface anomalies were used to guide our field research. Specifically, a GPR scan from the 2024 field season indicated a filled embayment and related features in a field to the east of the main settlement, These features became the targets for our investigation during the 2025 season. We opened standard archaeological excavation plots and trenches surrounding the subsurface embayment as a means to verify the existence and extent of key features. Two features, an arm to the embayment and the sidewall, were as expected when we trenched. Two others, the inlet and a hardpan feature, were not. At the northwestern corner near the hypothesized inlet, Trench M, our excavation revealed no lagoonal clays. Rather it revealed high quantities of shells interspersed with smooth gravel, indicative of a beach setting. Near the center, Trench R revealed we encountered a rocky linear feature 30-50 cm below the surface which was not evident on GPR scan. Finally, an intriguing circular feature, thought to be an ancient well was identified as a more modern feature. . Excavation yielded artifacts, both modern and ancient, mixed together indicating modern disturbance in that part of the site. While GPR is effective at locating significant features such as the lagoon, it often misses smaller ones.

This project is supported by the National Science Foundation.

25. No Abstract Assigned

26. Monitoring Mice Sexual Dimorphic Bisphenol A-induced DNA Damages by Modified Agarose Gel Electrophoresis

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DNA damage is a natural consequence of daily life. Various factors, originating from both environmental stresses and endogenous metabolism, pose a continuous threat to the integrity and functionality of the human DNA within the genome. If it left unrepaired can produce mutations, chromosomal aberrations, and induce cell death. Detection, characterization, and quantification of these genomic modifications and strand breaks provides important information about genotoxic exposures and cellular responses. These data are critical for health risk assessment, especially to monitor the effectiveness of chemo-preventives, or the efficacy of chemotherapies. There are several current techniques for detecting DNA damage and others for evaluating the underlying repair mechanisms. The limitations for those techniques need deliciated well-trained personal and high cost of instrumentations. The goal for this project is to develop the novel 3D-printed device to do pre-scanning and determination of chromosome DNA damages in low-cost and fast ways. Agarose gel electrophoresis (AGE) is a common molecular biology technique that utilizes an electric field to separate macromolecules by size, such as nuclear acids or proteins. To get the better resolution of AGE, it is critical to optimize the existing AGE running conditions. We examined the effects of voltage, percentages of agarose gel, staining conditions and destaining periods, as well as the concentration of chromosome DNA. As expected, increasing voltage subsequently increased electrical current, buffer temperature, and migration rates. A lower agarose gel percentage resulted in a higher resolution with genomic DNA samples, though a significantly more brittle gel. The DNA concentration of running samples directly corresponds to the intensity and size of the DNA bands. Loading samples of low DNA concentration require smaller amounts of sample, though increase the degree of uncertainty and risk of unquantifiable bands. Current AGE data we collected will be used to solidify the standard measurements and conversions for future sexual dimorphism projects in DNA damage detection.

This project was supported by the UROP/BUILD program, CSULB Mini-Grants and Summer Stipends (MGSS) Grant and KURE Incubator. We acknowledge the generous support of the WM Keck Foundation.

27. No Abstract Assigned

28. Racial and Health Equity in Long Beach City

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Equity in health, education, and economic opportunity remains a critical challenge for many urban communities. In Long Beach, racial and age disparities have been linked to gaps in educational attainment, poverty rates, and access to health care. This project in collaboration with the City of Long Beach Department of Health & Human Service, investigates racial and health equity challenges in Long Beach, California, by exploring the public use microdata sample (PUMS) data from the U.S. Census via the tidycensus package. The study focuses on how race, age, and socioeconomic variables influence disparities in education, poverty, health coverage, and income across North, East, Central, and South regions. Data wrangling involved narrowing down over 500 variables to equity-relevant measures, and public assistance participation. The final deliverables includes interactive Tableau dashboards and Infographics, aiming to make hidden equity patterns more visible, interpretable, and actionable for community stakeholders.

This project was supported by the Long Beach Compton Data Science Learning Community

29. Data-Driven Storytelling: Understanding Youth Needs Through Demographics and Arts Education in Compton

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Color Compton, a nonprofit arts and community archive, integrates creative expression and storytelling to empower Black and Brown youth in Compton. Even though the program plays an important role, it hasn't been easy to measure its full impact because the data is scattered and not well organized. This project investigates how arts-based programming

influences youth identity development, educational outcomes, and community-rooted self-perception by organizing and analyzing surveys, and demographic data.

We hypothesized that structured data analysis could reveal measurable impacts of Color Compton's programs on educational trajectories and civic engagement, particularly for students of color. Our team consolidated data from: American Community Survey (ACS) – 2010–2023 , California Department of Education (DataQuest) – 2018–2019, LA County Arts Ed Profile , and U.S. Department of Commerce . Using Python and SPSS, we applied exploratory data analysis, and descriptive statistics to assess participation demographics, geographic reach, and pathways into higher education and creative careers.

From our analysis, we discovered that in Compton Unified, that fine arts is not a high priority in their curriculum. Even more, there is an uneven distribution of opportunities that can shape which creative pathways students can realistically explore. Highlighting the displacement Compton has with their art programs and opportunities.

With these data-driven findings will help Color Compton better understand the surrounding communities, so they can design programs that directly respond to local needs. The insights also provide evidence to secure resources and build models that foster civic engagement, artistic growth, and self-advocacy. Overall, this work shows how community-focused data science can strengthen arts education and open clearer pathways for young people to higher education and creative careers.

Grant Support: This research is supported by the Global Hispanic Serving Institutions Equity Innovation Hub, California State University, Northridge.

30. Synthesis of Active Site Fragment Hits Targeting the EV-D68 RNA-Dependent RNA Polymerase

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2025 University of California San Francisco Summer Research Training Program

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Enterovirus D68 (EV-D68) is a non-polio picornavirus that has caused quasi-biennial outbreaks in the United States and Europe. It is primarily associated with severe respiratory illness in children and has been increasingly linked to neurologic complications, including acute flaccid myelitis. There are no approved vaccines or antiviral treatments for EV-D68. Given the virus's recurrent outbreaks and growing public health impact, there is an urgent need to develop novel therapeutics. The viral RNA-dependent RNA polymerase (RdRp/3D^{pol}) is essential for viral genome replication and serves as a highly conserved and druggable antiviral target. A recent crystallographic fragment screening campaign against EV-D68 3D^{pol} identified multiple ligandable sites, including five fragments bound within a conserved active

site, where a glycine-aspartate-aspartate (GDD) motif functions as the enzyme's catalytic center. We focused on the chemical synthesis of these active site-binding fragments using standard organic synthesis techniques. Structural fidelity of the synthesized compounds was confirmed by NMR and mass spectrometry. The lead fragment of interest (ligand x0304) forms a hydrogen bond with Asp325 of the GDD motif, a conserved catalytic residue essential for RNA chain elongation. This interaction suggests the fragment may interfere with catalysis and disrupt viral RNA replication. Additional fragment hits near the RNA exit channel engage in hydrogen bonding and Van der Waals interactions with RdRp, also supporting the potential to modulate RdRp enzymatic function through allosteric effects. Together, these efforts aim to enable the development of non-nucleoside inhibitors with high binding affinity and broad-spectrum antiviral activity against EV-D68 and related picornaviral RdRps.

This project is supported by UCSF Graduate Division.

31. Producing A Quadruple Mutant In Arabidopsis thaliana

Ruben Tapia, Anthony Aldana, Manrique Williams, and Judy Brusslan, Ph.D.

Leaf senescence (LS) of older leaves is a process that plants undergo after shifting from the vegetative state to the reproductive stage. During LS nutrients are remobilized from older leaves to new parts of the growing plant. Optimizing LS to the right time can help plants increase crop yield. Zinc Finger Homeodomain (*ZF-HD*) and Basic Leucine Zipper domain (*bZIP*) are two major gene families regulating transcription in *Arabidopsis thaliana*. Two bZIP genes (*bZIP34* and *bZIP61*) and two ZF-HD genes (*HB34* and *HB23*) were predicted to regulate bolting-induced LS (Hinckley and Brusslan, 2020). *-ZF-HD* genes can regulate flower development and timing, but information about paralogs *HB23* and *HB34* remains limited. *hb23hb34* lines displayed high chlorophyll content during bolting-induced LS, but NIT2 expression did not change, suggesting that *hb23hb34* lines showed a marginal response in LS compared to wild type (Michelle Smith, 2024, Thesis). Both *bZIP34* and *bZIP61* have been found to show altered expression in response to pathogens, light and stress signaling, and responses to phytohormones (Jakoby et al., 2002). They have been identified as bolting associated genes (BAGs, Hinckley and Brusslan, 2020). Similar to *hb23hb34*, *bzip34bzip61* lines displayed a small delay in bolting-induced LS, expressed via slightly reduced NIT2 and higher chlorophyll content; measured T0-T12. We hypothesized that generating a quadruple mutant will likely have an additive effect to the delay in leaf senescence. *hb23hb34* was crossed with *bzip34bzip61* and the heterozygous F1 was produced. F2 seeds were collected and screened for higher order mutants. The isolated DNA was used as a template for the polymerase chain reaction (PCR) to then subject the samples to gel electrophoresis to view PCR products. Each plant will have 16 possible gametes for the four genes. With 16 gametes from each plant a cross between 2 plants would lead to 256 possible combinations (1/16 x1/16). By limiting the type of F2 allele combination to double-mutant and double-heterozygous "double-double" we reduce the possible offspring allele combinations to 1 in

64. (¼ x ¼ x ½ x ½) = 1/64. We have confirmed that mutants *bzip34/61*and *hb23* were present in many of our samples. Our original HB34 primers amplified the HB23 paralog, and we are designing *HB34*-paralog specific primers. Generating a quadruple mutant may result in a significant delay in leaf senescence that may be helpful in increasing worldwide crop output.

32. Using an Agar-Based Protocol for Efficient Isolation of Diatom Viruses

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Marine viruses regulate phytoplankton populations including diatoms, single celled photosynthetic primary producers that produce roughly 20% of the world's oxygen. Isolating diatom-infecting viruses is important for understanding host-pathogen interactions, but few have been successfully isolated since their discovery in 2004. For eukaryotic phytoplankton, isolation experiments are typically conducted in liquid media using well plates due to the difficulty of growing these organisms on solid media. However, discerning successful infections in liquid culture is challenging, inefficient, and prone to false positives. Furthermore, multiple rounds of purification need to be performed to isolate viruses from liquid media, during which the target virus can be lost. Virus isolation using solid media allows viral infection to be easily identified as viruses form circular clearings, known as plaques, each of which are created by a single virus, simplifying the purification process. Here, we developed a protocol to grow diatoms on a solid medium using Low-Melting-Point Agarose (LMPA) at a low gel percentage of 0.35% that was amended with diatom-specific nutrients. We tested this protocol on 6 different strains representing diverse diatom lineages. Cultures that were typically <40 µm in diameter and that reached densities of at least 1 x 106 cells/mL were best suited for growth on LMPA plates. We then used this new protocol to isolate diatom viruses using samples from July 2024 to August 2025 from Los Alamitos Bay. Of the 89 LMPA plates made, 17 plaques were observed, giving a 19.10% detection rate. For comparison, liquid-culture challenges during the same period produced 171 clear wells across 93 well plates, averaging 1.8 clear wells per plate. The majority (94%) of these liquid challenges could not be further purified. While the LMPA reinfection experiments are still in progress, they will allow a direct comparison of efficiency between solid and liquid approaches in virus isolation. Establishing the most effective method will accelerate the isolation of novel diatom-infecting viruses and improve our understanding of the growth and death patterns of these important phytoplankton.

33. Investigating Spatial Regulation of Argonaute Protein Expression in the C. elegans Germline

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2025 University of Southern California Graduate School Jumpstart

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Khlifa S. Alnaim, M.S., Carolyn M. Phillips, Ph.D.

Gene regulation is vital for controlling cell function, ensuring proper development, and allowing cells to respond to environmental changes. One pathway that regulates gene expression is RNA interference (RNAi), a small RNA-guided mechanism first discovered in *C. elegans*. RNAi relies on a non-coding guide RNA that is loaded into an Argonaute protein to form a complex that targets complementary mRNA strands for downstream silencing. While some Argonaute proteins like WAGO-1 are expressed throughout the germline, others like ERGO-1 are restricted in their expression. Characterizing the regulators of Argonaute proteins is vital because RNAi is a key gene silencing pathway, and Argonaute dysfunction has been linked to neurodevelopmental disorders and immune disorders, while overexpression has been associated with certain cancers in humans. This project investigates how candidate regulatory genes influence the spatial expression of germline Argonautes using RNAi. We hypothesize that their knockdown will alter the expression and/or localization of ERGO-1 and NRDE-3. Four genes (GLD-1, FIP-1, PIE-1 and MEX-3) were knocked down via RNAi feeding. Genetically modified *C. elegans* strains expressing fluorescently tagged ERGO-1 and NRDE-3 Argonautes were imaged to assess changes in Argonaute expression following knockdown. L1 stage worms were placed on RNAi plates and imaged by fluorescence microscopy at defined time points: 68 hours to image the adult germline (when oogenesis occurs). The results from imaging revealed that certain genes such as GLD-1 and PIE-1 resulted in decreased expression and/or a more restrictive expression pattern. In contrast, knockdown of MEX-3 and FIP-1 did not have a significant effect on expression or restriction pattern. All and all, this project has shown that GLD-1 and PIE-1 play a role in regulating the expression of Argonaute proteins ERGO-1 and NRDE-3. which is critical for understanding conserved gene regulation mechanisms and may provide insight into diseases linked to Argonaute misregulation, such as cancer and immune disorders. These findings contribute to the ongoing research to map regulatory mechanisms of the RNAi pathway.

This project is supported by the National Institute of Health 5R35GM119656-10.

34. Functional Protein Characterization: Databases Linking Glycoside Hydrolases to their Activities and Substrates

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Glycoside Hydrolases (GH) are carbohydrate-active enzymes that degrade sugars. By enabling the breakdown of complex biomass, these proteins have broad scientific applications including human health, biofuel, biotech, ecosystem function, and agriculture. The functional annotation of GHs generally relies on "majority rule" where the predicted function of newly identified GH proteins corresponds to the most frequent activity of their GH family (e.g., new GH9s are "predicted cellulases"). However, as of March 2025, there were 234 unique activities unequally distributed across 162 GH families. To precisely predict the function of putative GH proteins, we introduced ez-CAZy and CAZy-sub. ez-CAZy is a custom reference annotation database linking characterized GH sequences to enzymatic activities. It was validated using 500+ recently characterized GHs – excluded from training – with high accuracy. After obtaining a predicted activity, identification of specific substrates for biochemical characterization may remain an obstacle. As of 2025, there are no clear guidelines—which substrates to test, how many, what methods to use—to identify GH activity(ies). Here, we compiled the list of substrates from BRENDA db for all the described activities from CAZy and provide a general path for defining which substrates should be tested on a putative GH enzyme. Together, these approaches will enable more precise prediction of GH activities and facilitate faster, more accurate characterization.

The material generated and analyzed in this study is based on work supported by the National Science Foundation under Award Number #2138880 (RB), the D.E. Gress Endowed Summer Research Fellowship (DE), and the Dr. Vern Eveland Memorial Award (DE).

35. Database Development and Approach for Annotating Unassembled Short-Read Sequences

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Microbe-facilitated carbohydrate degradation is critical to environmental processes (i.e. carbon cycling). Glycoside hydrolases (GHs) are the most abundant class of enzymes involved in microbial carbohydrate degradation, making them a topic of ecological

significance. To this end, metagenomic analyses are conducted to investigate the presence and distribution of GHs across environments. Current metagenomic analysis pipelines make use of genome assembly and metagenome-assembled genomes; however, these pipelines are limited by the complexity and coverage of the sample. To address this, we developed a reference annotation database to facilitate the identification of GHs within unassembled, short-read DNA sequences. Our FASTA database, $CAZyO_{GH}$, was constructed from Protein family (Pfam) profiles of 99 GH families. We hypothesize that identifications made from short reads are accurate and capable of profiling unassembled metagenome samples for GH content. $CAZyO_{GH}$ was validated prior to the analysis of novel unassembled data using simulated short-read sequences from 10,000 bacterial genomes. Short-read hits from $CAZyO_{GH}$ were then transformed via linear models and compared with gene counts from an additional subset of 1,000 bacterial genomes. From the validation, we observed that converted short-read hits are similar to full gene counts for the majority of GH families. Additionally, we applied this database to a case study using unassembled metagenomic data from soil samples collected on Santa Catalina Island, CA, USA. From short-read data, profiles of constituent GHs were generated and compared across samples. The results of the GH profile comparisons suggest a selection of bacterial communities capable of concurrent degradation of various substrates. The $CAZyO_{GH}$ reference database will provide a means for analyzing GHs within complex, unassembled metagenomic samples.

This work is supported by the National Science Foundation under Award Numbers #2138880 (RB).

36. Phylogeny and Environment Drive the Functional Potential of Glycoside Hydrolases across Microbial Lineages

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Glycoside hydrolases (GHs) break glycosidic bonds and play a key role in both microbial metabolism (e.g., energy production) and ecosystem function (e.g., carbon cycling). Previous genome analyses have revealed a phylogenetic constraint in GH distribution. However, most analyzed genomes were derived from pathogens (e.g., *Mycobacterium*) or from lineages selected for their biotechnological potential (e.g., *Escherichia*). Here, we examine the distribution and multidomain architecture of GHs from over 70 families across 52,515 metagenome-assembled genomes (MAGs). Specifically, the construction of MAGs from 10,450 environmental metagenomes offers an unbiased opportunity to link functional potential for polysaccharide deconstruction to specific lineages and environments. Additionally, we describe the precise multidomain architecture of the GHs (i.e., association with CBMs) and highlight lineage-specific adaptations for polysaccharide deconstruction. Beyond highlighting how phylogeny and environment shape microbial lineages, this work will

help identify novel lineages with high potential for polysaccharide deconstruction and new multidomain enzyme architectures.

This data generated and analyzed in this project is supported by the National Science Foundation under Award Numbers #2138880

37. Design and Development of a Transradial Prosthetic Device with Pressure Feedback for Manipulation Tasks

Amaris Guevara, Hector Mendoza, and Emil Demircan, Ph.D.

For years engineers have tried to create advanced upper-limb prosthetics that are efficient and affordable. Yet, a common reoccurring issue is not being able to give any sensory feedback to the patient which is important for motor learning. As a result most users are unaware of how much pressure they are applying to the item they grasp. The study's purpose is to create an upper- limb prosthetic with pressure feedback for the users to manipulate fragile objects. During this study, the design of a prosthetic will be developed and built off of SolidWorks. Mechanical actuation will be used and incorporated into the design. Space and size are boundaries needed to work within. Using mechanical actuation and force sensitive resistors for pressure feedback, we aim to create a working prosthetic with sensory characteristics. Quantitative and qualitative tests are performed as the design is tested. A draft is currently in progress.

38. Digital Light Processing (DLP)-mediated Control of Paracrine Interactions for Vascular Assembly

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Vasculogenesis, or the formation of new blood vessels, is a current challenge for the survival and function of organoids and engineered tissues, both in vitro and post-implantation. The lack of adequate vascularization prevents proper blood flow to deeper tissue, leading to hypoxia, nutrient deficiencies, and subsequent cell death. The use of stromal cells offer a dynamic and biologically responsive approach to promote vasculogenesis through the secretion of a diverse range of growth factors, cytokines, and extracellular matrix proteins. By leveraging the natural cues of stromal cells and allowing interaction with endothelial cells upon vessel formation, our approach aims to recapitulate the intricate cellular interactions and microenvironment present in vivo. We hypothesize that distal cues, caused by stromal factor diffusion, influence vessel formation in a high-throughput, multimaterial environment, thereby controlling vasculogenesis. 2.5 mg/mL fibrin gels and 7.5% gelatin-methacrylate

(GelMA) constructs were first evaluated for cell viability and diffusivity using FITC-dextran. Multi-material test prints assessed vasculogenesis based on distance and cell amount. To create these prints, human dermal fibroblasts (HDFs), a common stromal cell type in connective tissue, were encapsulated in 7.5% GelMA cubes containing 0.1% LAP via DLP-based bioprinting. 2.5 mg/mL fibrin gels containing encapsulated human umbilical vein endothelial cells (HUVECs) were added around the constructs. After 3 days, the prints were immunostained and imaged. HDFs led to significant vessel formation, suggesting that stromal factor diffusion plays a role. GelMA cubes without HDFs exhibited limited HUVEC capillary formation. GelMA prints spaced 2 mm apart promoted faster vessel formation in comparison to other distances. Future work focuses on further investigating the mechanical effects of vessel network creation. Collectively, this work contributes to the advancement of vascular assembly in engineering tissues through paracrine interactions and mechanical effects.

This research is made possible by the UC Irvine Summer Undergraduate Research Fellowship and the Smith Lab.

39. DREAM Regulated Gene Expression in Wild Type and Knockout Zebrafish Larvae Exposed to TBBPA

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Calcium (Ca^{2+}) signaling is needed for countless cellular functions and thus a wide range of physiology processes from egg fertilization to cardiac muscle contraction and neuronal communication. Unfortunately, environmental pollutants such as Tetrabromobisphenol A (TBBPA), disrupt cellular Ca^{2+} homeostasis by altering Ca^{2+} channels like the L-voltage gated Ca^{2+} channels and the ryanodine receptor. Few studies have examined how these disruptions alter Ca^{2+} regulated transcription. To address this data gap, our study focused on the Ca^{2+} sensor DREAM (Downstream Regulator Element Antagonistic Modulator), a transcriptional regulator that binds to DNA and regulates transcription in a Ca^{2+} dependent manner. Previous work in the Holland lab showed that DREAM regulated transcription is altered by TBBPA in neuroendocrine cells, but little work has been done in exposed organisms. Therefore, we assessed transcription in wildtype (WT) and DREAM knockout (KO) zebrafish models exposed to TBBPA for various time points. Specifically, zebrafish exposure started at 48 hours post fertilization and larvae were exposed for 24-, 48-, or 72 hours to an embryo water control, a vehicle (DMSO, 0.01%) or TBBPA at 0.73, 1.46, or 3.0 μM in 0.004% DMSO. After exposure, the larvae were pooled to provide enough material for RNA extraction

and then assessed for transcript levels of DREAM regulated genes *irs2a*, *insa*, *fosab*. Transcripts were normalized to the reference gene *efl1α* and then normalized to the respective DMSO transcript levels (ie. WT or KO). Preliminary results show that all genes in WT and KO larvae displayed time and concentration specific changes in expression. The exposed WT seemed to display the most drastic changes in expression after just 24 hours of exposure, especially for *irs2a* and *ins*. The KO displayed more moderate changes compared to the WT potentially suggesting decreased sensitivity to TBBPA induced transcriptional changes. However, this work will need to be repeated due to loss during exposures and RNA extractions leading to low sample sizes (n=1-6) and findings that DMSO may be affecting expression for the genes of interest. Overall, this work expands our knowledge on DREAM in zebrafish, and how pollutants may interrupt calcium mediated transcription important to cellular and organismal physiology.

Research supported by the KURE Incubator Award and the National Institutes of General Medicine (SC3 GM132033-01A1)

40. Cell Envelope Biogenesis in the Alpha-proteobacteria *Ochrobactrum anthropi*

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The cell envelope of Gram-negative bacteria is the main interface between pathogens and their hosts. It comprises the cytoplasmic membrane (IM), the peptidoglycan, and an asymmetric outer membrane (OM) containing phospholipids (PLs) in the inner leaflet and lipopolysaccharide (LPS) in the outer leaflet. Proper assembly of this complex structure is essential for bacterial viability and pathogenicity. However, the mechanisms governing envelope biogenesis and homeostasis in diderm bacteria remain partially understood. Previous work in the alphaproteobacteria *Brucella* suis demonstrated that MapB, the homolog of TamB from gammaproteobacteria, is critical for OM integrity. TamB is a large AsmA-like protein with a taco-shaped β sheet periplasmic domain anchored to the IM that was recently proposed to be involved in PL anterograde translocation in *E. coli*. In previous work, we demonstrated that the *ΔmapB* mutant displayed increased sensitivity to membrane-disrupting agents, defects in cell division, and impaired assembly of specific OMPs, supporting a broad role of MapB in envelope stability. To further investigate the role of the TAM system in Alphaproteobacteria, we generated and characterized *tamB* mutant (*ΔtamB*) in *Ochrobactrum anthropi* 49188 background, another gram negative bacteria that is the closest bacteria phylogenetically related to Brucella. Different approaches were employed to characterize the wt and *ΔtamB* in *Ochrobactrum*, including a lysozyme tolerance assay and growth curves at 37°C, 18°C and 12°C. Tolerance assays showed that

ΔtamB is selectively affected by lysozyme, indicating that the OM of *ΔtamB* is compromised, as lysozyme targets the peptidoglycan layer of the periplasm. The growth curve revealed a significant delay in the lag phase of growth of the *ΔtamB* at 12°C regarding wt and the complemented strain, revealing that TamB plays a role in adaptation of the cell membrane at cold temperatures, which entails changes that maintain membrane fluidity. In order to elucidate possible differences in PL composition between the IM and OM of the wt and mutant, cellular fractionation and lipid extraction were performed. Furthermore, in order to find possible functional interactions with the TAM system, we performed a suppressor mutant selection assay by incubating the mutant strain under restrictive conditions (12°C) and isolating spontaneous colonies that were able to grow. Candidate colonies were further incubated in liquid media at 12°C to verify that their growth rate was similar to the WT strain. Characterization of the cell membranes, as well as sequencing of the suppressor mutant candidates, is ongoing. Overall, our results indicate that TamB is involved in cell envelope homeostasis in Alphaproteobacteria, although further research is required to elucidate the underlying mechanisms.

Supported by the National Institutes of Health - Minority Health and Health Disparities Research Training (MHRT) Program (T37), Grant T37MD001368

41. Food Insecurity, Depression, and Health Risk Behaviors

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Food insecurity is limited or uncertain access to adequate and nutritious food in the United States and is linked to negative health outcomes such as mental health challenges and risk behaviors. Food insecurity is associated with increased engagement in health risk behaviors that translates into a heightened psychological distress which increases vulnerability to negative mental health issues and harmful behaviors; additionally, depression moderates this association by amplifying risk as food security worsens. Cross-sectional analysis of NHANES 2017-2018 using survey-weighted logistic regression for smoking and substance use and ordinal regression for alcohol use and depression, with models adjusted for income, age, education, and race-ethnicity, and additional models without covariates testing Food Security by Depression interactions via design-adjusted joint Wald tests, along with predicted probability plots. Relative to high food security, low and very low food security were associated with higher odds of smoking, substance use, heavier alcohol use, and more severe depression. Moreover, the Food Security by Depression interaction was statistically significant for smoking, drug use, and alcohol use, with predicted probabilities showing

steeper increases in smoking and drug use as food security worsens among those with moderate to severe depression, while at marginal food security the odds of substance use were lower than among those with high food security. Food insecurity is linked to adverse behavioral and mental health outcomes, and depression moderates these relationships, suggesting that policies and programs should incorporate food access support in mental and behavioral health services to reduce disparities.

This project is supported by the CSULB Center for Health Equity Research.

42. Qualitative Insights into Service Dog Training Guidelines: A Nationwide Study

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Post-traumatic stress disorder (PTSD) and traumatic brain injury (TBI) are prevalent in veterans and are frequently associated with delayed or long-term problems regarding health and well-being, and reintegration. Service dogs have been identified to treat PTSD symptoms, reduce sleep difficulty, and improve overall quality of life. However, with a lack of standard national training guidelines, programs have varied and may change the quality and effects of trauma. To address this gap, we employed a national mixed-methods study with 23 organizations that were part of the Wounded Warrior Service Dog Program. We employed the Delphi consensus method, developing approximately 100 training guidelines, which we distributed to subject matter experts to respond to in multiple rounds of surveys. Consensus was considered to be agreement in the study responses of ≥80% to define agreement. Quantitatively, we found that 91.3% of organizations trained in facilities; psychiatry needs were overwhelmingly most prevalent among service members (95.7%), while in the last 36 months, graduation percentages were reported at 79.1% for service members and 43.7% for service dogs. Qualitatively, the study findings emphasized the service dogs' welfare, the effectiveness of training, and satisfaction for veterans as priority areas.

This research provides a framework for national, evidence-informed through expert consensus and veteran experiences to bring consistency, safety, and outcomes to service dog programs. The standards have the potential to support well-established long-term relationships between veterans and service dogs, promote the durability of training organizations, and enhance the health and quality of life for veterans.

Funding Sources/Acknowledgments: This research was supported by the National Institute of General Medical Sciences of the National Institutes of Health under Award Numbers UL1GM118979, TL4GM118980, and RL5GM118978, with additional support from the U.S. Department of Defense Wounded Warrior Service Dog Program via the Uniformed Services University of the Health Sciences under MTEC Other Transaction Agreement W81XWH-15-9-0001. The content is solely the responsibility of the authors and does not necessarily

represent the official views of the NIH, DoD, or USU. The authors thank the BUILD program for mentorship and Dr. Jeremy Ramirez for his guidance.

43. Production of CRISPR/Cas9 Tools for the Deletion of C1q in Thp-1 cells

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C1q is a recognition protein of the innate immune system and is an essential subcomponent within the C1 complex of the classical complement cascade. In the absence of other complement proteins, *C1q* is a powerful immunoregulatory gene that aids in the inflammatory response. This has been previously demonstrated in *C1q* knock out mice in which impaired apoptotic cell clearance and autoimmune disease susceptibility has been observed. Although these studies suggest that *C1q* has a key role in macrophage autophagy, the full extent of the intracellular pathways involved in C1q-macrophage interactions remains to be fully understood. What could potentially provide further insight would be the production of *C1q* deletions in human cells. Therefore, we hypothesized that CRISPR/Cas9 technology, a highly effective genome editing tool, can be used to produce a gene deletion of *C1q* in Thp-1 macrophage cells. Specifically, through CRISPR/Cas9 multiplexing, two unique single guide RNAs (sgRNAs) that are specific to the *C1q* gene will be used to direct the Cas9 endonuclease to introduce double strand breaks (DSBs) at two different sites. After cleavage, the genomic DNA will repair itself via the endogenous nonhomologous end joining pathway (NHEJ) resulting in insertions and deletions (INDELs) at the site of the DSB, effectively knocking out the gene. These modified cells will then be used to further confirm that *C1q* is highly involved in the inflammatory response while also advancing our understanding of the intracellular mechanisms of *C1q*. Six single guide RNAs were identified that would direct the Cas9 protein to either the start or end of the *C1q* gene. These sgRNAs were ordered as single stranded oligonucleotides, annealed together to form a double-stranded molecule with the appropriate overhangs and then phosphorylated for insertion into a Cas9 expression plasmid. The circular Cas9 expression plasmid, referred to as pX330, was linearized with the *BbsI* restriction enzyme. Each of the prepared sgRNA oligonucleotide strands were then inserted (ligated) into a linearized pX330 plasmid. All potential ligation products were separated into bacterial cells and grown as individual colonies on LB agar plates with ampicillin resistance. The plasmid DNA was isolated from individual colonies and sequenced to confirm sgRNA integration. The resulting six sgRNA containing pX330 plasmids were then delivered to Thp-1 cells via nucleofection. Delivery of exogenous plasmid DNA was confirmed using a GFP expression plasmid as a positive control. Genomic DNA will be evaluated for the presence of INDELs, and the most efficient sgRNAs as assessed by the greatest amount of INDELs will be used to perform *C1q* gene deletion in Thp-1 cells. The effects of *C1q* absence will be confirmed by PCR and sequencing, and the

gene deleted Thp-1 cells will be analyzed to assist in our understanding of C1q-macrophage interactions.

This project is supported by the National Institutes of Health under Grant 1R16GM149507.

44. Optimization of sgRNAs for CRISPR/Cas9 Editing of the RhD Locus and Insertion into a Cas9 Expression Plasmid

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Blood types are determined by the presence or absence of proteins on the surface of red blood cells. One of these proteins, the Rh factor, determines if an individual has a negative or positive blood type. Specifically, the *RhD* locus is disrupted in individuals with negative blood types. However, a phenotypic consequence has not been observed to be associated with the disruption of this gene. Therefore, the *RhD* locus is being evaluated in the Johnston lab as a potential safe harbor location for CRISPR/Cas9 mediated gene addition.

Previously, single guide RNAs (sgRNAs) were manually identified near the start codon of the *RhD* locus that could be used to guide Cas9 to the *RhD* locus to produce a double strand break in the genome. However, the sgRNAs produced showed limited cutting efficiency when examined *in vitro*. We hypothesize that alternative sgRNAs within the RhD locus could be identified and produce greater cutting efficiencies.

Bioinformatic tools were used to identify sgRNAs with greater cutting efficiencies and minimal off-target effects than those previously produced. Seven sgRNAs with high specificity were ordered as oligonucleotides and incorporated into Cas9 expression plasmids. Integration for two of the sgRNAs was confirmed by sequencing. Confirmation of the other five sgRNAs is in progress. Once all the recombinant plasmids have been produced, we will deliver these plasmids to a human cell line: (1) a myeloid erythroid K562 cell line. Delivery will be accomplished using electroporation which temporarily increases the permeability of plasma membranes allowing for entry of exogenous DNA into the nucleus. Seventy-two hours after electroporation, the genomic DNA will be evaluated for insertions and deletions (INDELs) which occur as a result of the endogenous non homologous end joining repair pathway following a Cas9 induced double strand break. The sgRNA that results in the most INDELS will be deemed the most efficient and used to perform subsequent gene editing experiments.

The project was supported in part by a Kure Incubator award.

45. CRISPR/Cas9 Editing of Phaeodactylum tricornutum

Rylee P. Ortivez, Darla G. De Leon, Jason Diaz, Ella P. Morris, Michael C.G. Carlson, and Jennifer M. Johnston

Viruses are the most abundant biological entities on the planet. The ability to produce viruses in a test tube is important in for their characterization and analysis. Currently, some of the only methods available to grow viruses in the lab limit the diversity of viruses able to be isolated because they are time-intensive and depend on the selectivity of the host. To combat this problem, we will genetically engineer a host cell, using the CRISPR/Cas9 system to produce a virus. Using the diatom *Phaeodactylum tricornutum*, an essential oceanic oxygen producer, as a model system, we plan to insert a diatom infecting Bacilladnavirus genome into the diatom genome. We will achieve this by inserting two pieces of recombinant DNA in the form of plasmids into our diatoms through electroporation. Here we optimized the ratio of DNA to pulses from the electroporator for the diatoms to confirm that we can consistently insert plasmid DNA containing GFP into *P. tricornutum*. Transformed diatoms fluoresced more green than compared to the (non-transformed) wild type *P. tricornutum*. The next step is to deliver the CRISPR/Cas9 system contained on one plasmid and the DNA repair template containing the viral genome on a second plasmid into the diatoms. To do this, we first needed to build these plasmids. We identified several single guide RNAs (sgRNAs) that would be able to direct the Cas9 protein to a intergenic location of the *P. tricornutum* genome. These sgRNAs were ordered as oligos and incorporated into a Cas9 expression plasmid as confirmed by sequencing. We are currently in the process of building our DNA repair template that will be used by the diatom to repair the double strand break induced by Cas9. Once these components are made, we will use the electroporation protocol we optimized to introduce the CRISPR/Cas9 system and DNA repair template into *P. tricornutum* and study the virus produced.

46. Tracking Diatom–Virus Interactions: Isolation and Ecological Dynamics in Southern California Current Ecosystem

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Diatoms are photosynthetic unicellular microbes responsible for approximately 40% marine primary production and are infected by viruses that are suspected to control their abundances in the oceans. Understanding the ecological and physiological choreography between diatoms and their viruses is thus important, but a limited number have been isolated and characterized since their discovery in 2004. We aim to expand the catalogue of viruses in culture and to understand their genomic and physiological characteristics. To accomplish this, diatom cultures were challenged with environmental virus samples

collected regularly from June 2024 to July 2025 from Los Alamitos Bay, Long Beach and Newport Bay wetlands in Southern California. We tested several approaches to optimize virus isolation, including using 41 different diatom strains, employing both seawater and sediment sources, challenges in both liquid and solid media, and varying host-to-virus ratios. For these challenges, control and virus-challenged cultures were monitored for changes in RGB-luminosity, chlorophyll fluorescence, and lawn homogeneity, as well as microscopically for cell wall rupture or chloroplast shrinkage. Out of 2045 virus-diatom challenges, 188 challenges (9.2%) demonstrated diatom cell lysis and decreased host abundances relative to controls. The diatoms *Chaetoceros*, *Skeletonema*, and *Thalassiosira* demonstrated the highest percentage of successful challenges (11.0%, 10.3%, and 8.6%, respectively). Samples collected after the decline of diatom bloom in October 2024 caused the highest infection percentage (11.8%). This suggests that viruses closely follow diatom abundances. Seawater and sediment samples yielded similar infection percentages (9.1% and 10.4%), indicating both are reservoirs for infectious viruses. Of the 188 viral fractions from these putatively lysed cultures, six reproducibly caused diatom cell death across three successive reinoculations into healthy cultures. Three of these six lysates were purified through dilution to extinction to isolate a single virus, suggesting we have cultured three new *Skeletonema*-infecting viruses. These new diatom-virus cocultures pave the way to understanding the genomic properties of diatom-infecting viruses and the molecular and physiological mechanisms underlying diatom-virus interactions.

47. Isolation of Diatom Infecting Viruses in Southern California and Their Ecological Significance

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Diatoms are unicellular, microscopic algae encased in silica shells that play a crucial role in marine ecosystems. They are responsible for producing approximately 20% of Earth's oxygen, form the foundation of marine food webs, and are key contributors to nutrient cycling in the ocean. Diatom populations are susceptible to infection by viruses, which can disrupt blooms and drive significant ecological change. To date, only about twenty diatom-infecting viruses have been isolated, emphasizing the need for further research into their impact on diatom mortality and their broader ecological significance. The primary goals of my research are to isolate diatom-infecting viruses, assess their effects on diatom mortality, and determine their ecological roles across different diatom host genera in the ocean. Isolation involves combining cultures of single diatom strains with the virus-containing fraction of seawater and monitoring for signs of infection. Thus, optimizing the ratio of diatom culture to virus treatment is expected to improve infection efficiency. To identify the most effective method for virus isolation, 10% and 50% virus concentrations were compared

to determine which treatment yielded more putative infections. This study operates under the hypothesis that the 50% virus concentration will yield more putative infections because higher viral abundances may increase the probability of infective diatom–virus interactions. To examine these dynamics, exponentially growing diatom cultures were exposed to either 10% or 50% virus concentrations and monitored for changes in fluorescence. Wells or tubes showing reduced growth or signs of lysis relative to uninoculated controls were considered putative infections. The lysates from these dead cultures were then used to reinfect healthy cultures. Higher viral concentrations (50%) led to more successful initial infections compared to lower concentrations (10%). On average, 17.6% of trials at the 50% concentration resulted in infection, whereas only 10.6% of trials were successful at the 10% concentration. Reinfection patterns differed between hosts, with *Chaetoceros* reinfecting at 10% and *Pseudo-nitzschia* at 50%. Reinfected cultures exhibited decreasing fluorescence, supporting the Koch's postulates for the isolation of a putative virus. This research advances virus isolation techniques and deepens our understanding of how viral infections shape diatom dynamics and support biodiversity.

This project is supported in part by the California State University STEM Pathways Alliance.

48. Single-cell detection of cryptic viral and parasite infections in marine diatoms

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Diatoms are critical to global ecosystem stability, producing nearly 20% of Earth's oxygen and are major components of the carbon cycle through their contribution to global carbon fixation and carbon export via the biological carbon pump. Their population dynamics are influenced by both parasitic and viral infections, yet these interactions remain difficult to observe and quantify due to their microscopic size, taxonomic diversity, and limited morphological distinctiveness of both hosts and pathogens. This project explores the use of single-cell PCR (scPCR) as a new approach to directly detect parasite and viral infections in individual diatom cells. This method allows for precise genetic identification of infection events that can otherwise be morphologically undetected. We hypothesize that this approach will not only confirm the presence of infection but also reveal which pathogens are capable of infecting diatoms. We employed a nested PCR approach using degenerate, lineage-encompassing primers targeting the 18S rRNA gene of diatoms, the 18S-28S rRNA operon of parasites (*Chyrtidomycota*, *Oomycota*, and *Cercozoa*), and the viral replication genes of diatom-infecting viruses (*Bacilladnaviridae* and *Marnaviridae*). This method was tested on diatom cultures *Chaetoceros* and *Coscinodiscus*, which demonstrated successful amplification of DNA from individual diatoms, confirming the feasibility of amplifying minimal amounts of genetic material. We have also confirmed the ability of our new

pathogen-specific primers to detect diatom-infecting viruses and parasites. These results establish the technical foundation for the ability to duplex scPCR assays, in which pathogens and diatoms are detected simultaneously, indicating the presence of an infected cell. When applied to multiple cells from an environmental sample, this approach can provide quantitative estimates of infection levels and new insight into ecological role of parasites and viruses in regulating diatom dynamics. Ultimately, this work will inform predictions of how diatom–pathogen interactions may shift under changing ocean conditions, with broad implications for oceanic primary productivity and global biogeochemical cycling.

This project is supported by the COAST undergraduate research award, a COAST Grant Development Program Award, and KURE award.

49. What's the point?: barb morphology of the Round Stingray (*Urobatis halleri*)

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Round stingrays (*Urobatis halleri*) are commonly found along the Southern California coast. Many beachgoers are injured each year during encounters with round stingrays due to stingray strikes. The severity of injury is influenced by various factors, one of which may be due to the differing morphological features of the venomous barb, located on the tail of the stingray. The location of the barb and the morphological characteristics of the barb vary across stingray species; however, detailed studies are lacking on how certain morphological characteristics may vary within and across stingray species. We quantified n=12 morphological features from n=28 barbs from various individuals of the round stingray. Barb features included tip sharpness, thickness, second moment of area, and the number of lateral serrations. We collected round stingrays from Seal Beach, CA, USA, where their barbs were clipped before releasing the stingrays back into the ocean. The barbs were micro-CT scanned, and using the program Slicer, we compared the 3D morphological features across the different barbs. We found there was variation across all of the characteristics measured. We aim to expand our work by exploring how barb morphologies may change throughout ontogeny and between sexes. We also plan to compare barb morphologies beyond the round stingray and conduct interspecific comparisons of stingray barbs to better understand the phylogenetic relationships of stingray barb uses.

This project is supported in part by the Department of Biological Sciences and CNSM at CSULB.

50. Buried Alive: Sand Burial Behaviors of the Round Stingray (*Urobatis halleri*)

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Round stingrays (*Urobatis halleri*) are commonly occurring along Southern California's coastlines in nearshore sandy habitats. We previously observed that the rays will bury themselves by fluidizing sand from beneath their bodies, settling on their dorsal surface in a highly controlled manner. The rays appeared to simultaneously control the motions of their snout, pectoral fins, and tail to achieve this precise relocation of the sand particles. Our question addressed how the round stingrays bury themselves beneath the sand. We used n=8 rays (n=4 males and n=4 females) of various sizes to observe how sand burial behavior occurs across ontogeny and between sexes. Rays were collected at Seal Beach, CA, and we collected sand in the surf zone where the rays live. Two high-speed Edgertronic cameras were used to film the sand burial, a voluntary behavior, in both the lateral and dorsal views at 250 fps for 4.5 seconds. Our videos were imported to MATLAB software where we tracked the kinematics of the snout, pectoral fin, and tail motion in 3D for the entire sand burial behavior. We found the sand burial behavior to be a highly controlled set of simultaneous movements characterized by the undulation of the pectoral fins, a shovel-like motion of the snout, and oscillation of the tail. The stingrays displayed a well-coordinated set of motor control systems, most likely used to bury themselves for camouflage from predators and prey. Understanding the mechanics of sand burial behavior in round stingrays will allow for comparison with other burying organisms.

This project is supported in part by the Department of Biological Sciences and CNSM at CSULB.

51. Surf's Up: Protection of Stingray-Resistant Neoprene Surf Booty

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Round stingrays (*Urobatis halleri*) commonly occur along the coastlines of Southern California in nearshore sandy habitats. As a defense mechanism, round stingrays will strike their tails to impale their target subject with their needle-like barb. Humans wading in these nearshore habitats are often targeted by stingrays due to accidentally stepping on these benthic creatures. To reduce beach injury caused by round stingrays, DragonSkin® surf booties were created to protect those people who frequent the beach. The surf booty is comprised of two layers: an inner sock that rests inside a more rigid outer layer with differing regions of stingray-resistant material to allow for foot mobility. To investigate the total surface area and percent protection of stingray-resistant neoprene, an Artec Space Spider 3D scanner was used to scan both the inner and outer layers of the booty, creating digital

models of various booty sizes. Using CAD (computer-aided-design) software, the digital scans were aligned and detailed to calculate the surface area coverage of stingray-resistant material, offering a total percent protection against stingray strikes to individuals wearing the booty. The results will contribute to improving the overall design of the surf booty and to help reduce the risk from potential stingray strike injuries.

This project is supported in part by the Department of Biological Sciences and CNSM at CSULB.

52. Isolation of Diatom Infecting Viruses from Ocean and Sediment Near Long Beach, California

Kassandra Gutierrez and Michael Carlson

Diatoms are photosynthetic unicellular autotrophs which serve as the foundation of productive marine food webs, contributing to approximately 40% of marine primary production. They are infected by viruses that are important for controlling their populations. As a result, of viral infection, silica-ballasted diatoms are suspected to sink and accumulate at the bottom of the ocean, suggesting that marine sediments may be a likely place diatom viruses accumulate. This research aims to test whether seawater or sediment is more effective as a reservoir for viruses to enhance diatom virus isolation and characterization. To evaluate this, seawater and sediment samples were taken from Ballast Point in Long Beach and the wetlands in Newport Beach, California, respectively. The virus-containing fraction (<0.2µm) of these samples was inoculated to exponentially growing diatom cultures. Cell cultures were then monitored for signs of death relative to controls by observing changes in culture cell density, color, and fluorescence. A total of 416 seawater trials and 221 sediment trials were conducted, using diatoms from 3 genera including *Asterionellopsis* sp. (strain CSULB 2315), *Chaetoceros* sp. strains CSULB 2396 and 2414, and *Skeletonema* sp. strains CSULB 2340, 2354, and 2369. The fraction of seawater and sediment samples that displayed potential infection was 7.45% and 10.41% respectively. Samples of seawater generally showed signs of decline such as color loss or cell density loss in a shorter time, averaging approximately 6 days compared to sediment samples where the average was 9 days. Collectively, these data suggest that diatom viruses are likely abundant and remain infectious in marine sediments, This highlights the role of sediment as a stable reservoir for viruses and highlights the value of targeting sediments to improve isolation and characterization.

53. Characterizing Genetic Variants in Whole Genome Sequencing in Epithelial Ovarian Cancer Cases

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Epithelial ovarian cancer (EOC) is the most lethal gynecologic cancer, with most patients diagnosed at a late stage, with long-term survival below 30%. Inherited risk is well established, yet current clinical testing, focused largely on *BRCA1*, *BRCA2*, and a handful of other DNA repair genes, accounts for less than half of the heritable component of disease. This gap suggests that additional classes of genetic variation, particularly structural variants (SVs), remain underexplored contributors to risk. In this project, we examined germline SVs alongside pathogenic single-nucleotide variants in order to evaluate their role in EOC susceptibility in downstream analyses. For SV burden analysis, whole-genome sequencing data from ovarian cancer cases and controls were processed with DRAGEN (Illumina, CA) to identify deletions, duplications, and inversions in the genome. The germline SV burden (per megabase) was similar across cases and controls. Deletions consistently spanned larger genomic regions than duplications, highlighting potential loci of biological relevance. In parallel, we used the Illumina Global Diversity Array to profile EOC cases and annotated variants with ClinVar and dbSNP. Pathogenic variants were overwhelmingly concentrated in BRCA1 and BRCA2, which carry a known pathogenic deleterious variant. Excluding these carriers yields a refined non-carrier cohort for novel germline SV discovery. Taken together, these findings underscore both the dominance of BRCA genes in clinical testing and the likely contribution of less-characterized structural variants to the missing heritability of ovarian cancer. This study lays the groundwork for the discovery of novel non-BRCA germline risk alleles.

This research was supported by Cedars-Sinai's Cancer Research Training & Education Coordination Office (CRTEC) and the Undergraduates Gaining Research Opportunities for the Cancer Workforce (U-GROW) program.

54. Enrichment and Isolation of PAH-degrading Bacteria from Oil-impacted Southern California Salt Marsh

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Polycyclic aromatic hydrocarbons (PAHs) are hazardous pollutants released into the environment via the incomplete combustion or degradation of organic compounds as a result of both natural and anthropogenic activity. A common source of PAH pollution is the

burning, or spillage, of oil. An oil spill in 2021 contaminated the wetlands along the coast of Huntington Beach in southern California. Previous studies have shown microbes play a role in oil spill remediation. We hypothesized that PAH-degrading bacteria would be present in the Huntington Beach Wetland (HBW). HBW sediments were used to create enrichment cultures with phenanthrene (Phe), pyrene (Pyr), or benzo[ghi]perylene (BghiP) as exclusive carbon sources. Twenty-five strains were isolated via streak plating, eighteen isolates from seven distinct bacterial genera have been successfully identified through 16S rRNA gene sequencing. Whole genome sequencing of representative members of each genus will provide deeper insight into the biochemical mechanisms used by the stains to degrade these PAH compounds. PAH-degrading bacteria are potential candidates for future bioremediation efforts.

This project is supported in part by the Asian American and Native American Pacific Islander Student Success Center and Development Program (ASCEND) grant number P031C220040, and the California State University Council on Ocean Affairs, Science & Technology (CSU COAST) under the 2024-25 Undergraduate Student Research Support Program (to AT).

55. Role of Androgen Receptor in Mouse Home-cage Behavior from Young Adulthood to Middle Age

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Using an automatic home-cage monitoring system, Bernstein et al. (2024) observed that female mice were more active than males for the dark phase. In the current study, we test the hypothesis that the masculinization of home-cage behaviors was mediated by the androgen receptor (AR), which was abolished by the AR mutation or an age-dependent decline in hormones. We individually caged and video-recorded young and middle-aged, wild-type (WT) female, WT male, and testicular-feminized (Tfm) male C57BL/6J mice (n = 11-14 per group) in a Noldus' PhenoTyper with a 12h:12h light:dark cycle (light on 0600 h) for 54 h. The distance traveled, number of running-wheel revolutions, number of water licking, and number of feeding during a selected 24-hour period (from 0600 h on the second day to 0600 h on the third day) were measured by EthoVision XT. Regardless of genotype or age, all mice displayed higher activities of locomotion, wheel running, drinking, and feeding during the dark phase than the light phase. During the dark phase, Tfm males traveled less distance than WT mice regardless of age. In addition, young WT females displayed the highest number of wheel revolutions, followed by WT males, with Tfm males showing the lowest. Wheel running activities were reduced in middle age, and the sex differences disappeared. Besides locomotion, young wild-type females licked the water more than males during the dark phase while the numbers of licks in middle-aged mice decreased compared to young conspecifics. Together, our results demonstrate that in a home-cage setting, sex

differences, phase of the light cycle, aging, and functional AR modulate locomotion, drinking, and feeding in mice.

56. Androgenic Regulation of Sex Differences in Social Behavior of Young and Middle-aged Mice

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Androgens play a critical role in the organization and activation of sexually dimorphic social behaviors through the activation of androgen receptor (AR). As age advances, the decline in the production of androgens may be an important factor contributing to behavioral changes in males. To test this, we used computer-assisted video tracking to assess sociability and social novelty preference in young and middle-aged male C57BL/6J mice carrying the testicular feminization mutation (Tfm) as well as their wild-type (WT) male and female counterparts. Each subject was first scored based on time investigating a non-familiar female conspecific confined in a wire-mesh corral vs. an empty one for 15 min (sociability). After a break, the same subject was then tested by measuring exploration of the first, now-familiar, female vs. a second non-familiar female for 15 min (social novelty). In both behavioral tests, all young mice displayed a stronger preference for social over non-social stimuli throughout the entire 15-minute period and novel over familiar stimuli during the first 5-minute period, respectively. WT males spent more time investigating a female stimulus than WT females and Tfm males, while the numbers of visits to social and non-social stimuli were higher in WT females than WT and mutant males. In addition, a greater number of visits to a familiar stimulus and increased time spent on a novel stimulus were observed in WT females and WT males, respectively, than in their WT and mutated conspecifics. In summary, sociability and social novelty in mice are sexually dimorphic, and specific behavioral components may be differentially regulated by AR.

57. How does the Testicular Feminization Mutation Affect Mice Genetically and Phenotypically?

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Mice carrying the testicular feminization (Tfm) mutation serve as an ideal model for studying the role of the androgen receptor (*Ar*) in sexual differentiation of behavior and brain function. The goal of our project was to explore the fundamentals of this model and the research tools used to investigate brain sexual differentiation. Through this work, we aimed to gain a better understanding of the relationship between genotype and phenotype, and how the Tfm mutation influences both physical traits and behavioral outcomes in mice. The *Tfm* mutation

arises from a single base pair deletion in the *Ar* gene, resulting in an insertion of an early stop codon in mRNA, leading to a truncated AR protein that lacks both the ligand- and DNA-binding domains. The presence of the Tfm mutation can be verified by genotyping. We first conducted PCR to amplify a specific *Ar* DNA sequence from the tail DNA of wild-type and Tfm mice, followed by MwoI digestion. After gel electrophoresis and imaging, wild-type mice would show 2 small fragments (145 and 65 bp) while Tfm mice had only a bigger fragment (211 bp). In addition, to determine genetic sex, we amplified *Ssty1*, a Y chromosome gene, and *Myog* using PCR, followed by electrophoresis. Males, not females, were identified with the presence of *Ssty1*. In addition, we bred the heterogeneous females, carrying one Tfm and one wild-type allele of *Ar*, with wild-type males. By examining the presence of the mammary glands of their pups (n=13), we found that 76.9% phenotypically female. Genotyping revealed nearly equal distribution among four genotypes: 23.1% wild-type males, 23.1% wild-type females, 23.1% heterozygous females, and 30.8% Tfm males. Together, a 1/3 of phenotypically female individuals were identified as Tfm males. We participated in euthanizing three mice and learned to measure anogenital distance (AGD), identify mammary glands, and collect peripheral organs. Lastly, we analyzed social behavior in 13 mice by comparing the time spent and number of visits to an empty cup (non-social stimulus) versus a cup containing a stimulus female (social stimulus). Mice spent significantly more time ($p<0.0001$) and made more visits to ($p=0.0002$) the social stimulus compared to the non-social stimulus. In this study, we learned and applied a range of fundamental research techniques to determine the phenotypes and genotypes of wild-type and Tfm mice. Through hands-on experience—including anatomical assessments, genotyping, and behavioral analysis—we developed a deeper understanding of the biological consequences of the Tfm mutation.

This project is supported by the KURE Summer Bridge Program.

58. Bee- and Hummingbird-Pollinated *Justicia* (Acanthaceae) Produce Less Pollen than Lepidopteran- and Wasp-Pollinated Species

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Flower gamete production is closely tied to reproductive efficiency in angiosperms. In mutualistic relationships between plants and animal pollinators, flowers provide nectar or pollen as a resource, while pollinators transfer pollen to fertilize ovules. With a broad distribution across tropical and subtropical regions and extensive diversity in flower morphology, *Justicia* (Acanthaceae) is a valuable system for examining how floral traits and pollination strategies influence reproductive breeding systems. Corolla colors in this group range from red to pink to white, while tubes vary in length and width. They consistently have four ovules and two anthers, usually with two thecae each, that can vary in size, position,

and presence of extensions from the theca base. These traits have been shown to be closely associated with pollinator types, making *Justicia* an ideal system for exploring how selection by different pollinators can affect male gamete production. We investigated the relationship between pollinator type and Pollen: Ovule (P:O) ratio, an indicator of reproductive efficiency, across 28 species of *Justicia*. We counted the pollen grains in each theca and ovule number to calculate P:O ratios and we measured inner anther extension length to evaluate its influence on pollen quantity. Our results show that bee- and hummingbird-pollinated Justicia species produce significantly more pollen than lepidopteran- and wasp-pollinated species. Species with high P:Os, such as bee-pollinated *J. betonica*, must produce more pollen to accomplish successful fertilization, while species with low P:Os may be more efficient at reproduction. Bee-pollinated *Justicia* species have significantly longer extensions on their inner thecae than *Justicia* visited by hummingbirds or lepidoptera. Variation in P:O ratios and anther morphology reflect trade-offs between reproductive investment and pollination efficiency, offering insight into the evolution of reproductive strategies in flowering plants.

This work is supported by a CSULB Research Foundation Summer Student Research Assistantship to Kelsey Imrie in 2025, NIH RISE (5R25GM071638-16) funding, and NSF funding (#1754792) to Amanda Fisher.

59. Cool Lizards in Hot Cities: Urban Influence on Western Fence Lizard Eco-physiology

Bree Putman, Brenden C. Saraf

Urban heat islands, where urban core areas are warmer than surrounding natural areas, alter temperature and water availability in ways that may affect ectotherm physiology and behavior. Because urban environments create novel ecological challenges for wildlife, understanding how these stressors interact is crucial for predicting species persistence in cities. To investigate, we are comparing Western Fence Lizards (*Sceloporus occidentalis*), one of California's most abundant native vertebrates, across Southern California urban college campuses against those found in adjacent natural habitats. Our study quantifies abiotic conditions (temperature, humidity, soil moisture) and links them to lizard body temperatures, hydration state, and cutaneous evaporative water loss. We hypothesize that urban environments may provide buffer against urban core heat, through landscaping and irrigation, leading to higher heat tolerance through higher hydration levels. Ongoing work measuring Western Fence Lizard physiology will test whether hydration and thermal tolerance differ between urban campus areas and their adjacent non-urban sites. Preliminary results suggest that climatic differences between sites, and lizards found at on-campus urban sites maintain higher hydration. This project not only provides novel insights into urban eco-physiology, and whether prevalent urban water systems will have a positive effect on populations of native ectotherms but also fosters undergraduate research training and multi-campus collaboration.

Acknowledgements: California State University Long Beach, San Diego State University, Scripps College, Cal Poly SLO, UCLA, California State University San Bernadino, Harvey-Mudd University, Pepperdine University, California Lutheran University, California State University San Marcos, Whittier College

Funding: STEM-Net, RESCUE-Net, National Science Foundation

60. Innate Immune Protein C1q Induces Autophagy in Macrophage Foam Cells

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Complement protein C1q is known to opsonize modified low-density lipoprotein (LDL) particles, enhancing phagocytosis by macrophages. Macrophages that engulf large amounts of LDL store excess cholesterol in lipid droplets, forming foam cells. Previous research has supported additional roles of C1q in modulating the metabolism and degradation of LDL. Here, autophagy reporter cells are used to validate previous data that suggests C1q induces autophagy in foam cells. RAW-Difluo mLC3 reporter cells were treated with oxidized LDL with or without C1q, and autophagy was measured by fluorescence confocal microscopy. LC3 positive fluorescence intensity was quantified 2 and 24 hours after treatment. Preliminary data show that C1q increases autophagy. The ability of C1q to induce autophagy may improve survival in macrophage foam cells. This may be important in slowing or preventing the development of arterial plaques in atherosclerosis.

Acknowledgements: This research was supported by the National Institute of General Medical Sciences of the National Institutes of Health under Award Numbers 1R16GM149507. The content is solely the responsibility of the authors and does not necessarily represent the official views of the National Institutes of Health.

WP is supported by the CSULB BUILD Program in the Office of Undergraduate Research Services.

61. The Effect of *Mbnl2* Knockout on Sleep Architecture

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The etiology of Myotonic Dystrophy Type 1 (DM1) is closely tied to the loss of the muscleblind-like family of proteins (*Mbnl*), which are RNA-binding proteins that regulate alternative splicing pathways. Loss of *Mbnl1* leads to physical symptoms like muscle atrophy, myotonia, and immobility. Its less-studied paralog, *Mbnl2*, performs similar functions in the CNS, and its loss results in memory, cognition, sleep, and circadian rhythm deficits. We used an *Mbnl2* KO mouse model of DM1 to examine sleep architecture and sleep homeostasis. To do this, we employed polysomnographic electroencephalography (EEG) to assess baseline sleep over a 24-hour period. Our preliminary results showed

elevated REM sleep over total time slept. In addition to reduced sleep in *Mbnl2* KO mice, which indicates hypersomnia. We observed reduced wakefulness and increased NREM in *Mbnl2* KO mice compared to WT mice, specifically throughout the light or rest phase (ZT 0-12). In contrast, there were nominal changes in NREM sleep during the dark or active phase in *Mbnl2* KO mice. We also observed elevated REM sleep in both light and dark phases, but especially in the light phase. Although limited by low sample size, these results suggest that *Mbnl2* has an impact on sleep architecture.

62. Using PCMT1 to Identify Sites of Isoaspartylation in ELAVL4

Diego De Luna Moran, Shiori Fukutome, Diego A.Velarde , Ite A.Offringa

Small cell lung cancer (SCLC) is a highly aggressive disease with poor survival outcomes and limited therapeutic options. Approximately 15–20% of patients develop antibodies against ELAVL4, a protein expressed in pulmonary neuroendocrine cells, which correlates with improved response to chemotherapy and survival. Prior studies in the Offringa lab identified a 38-amino acid N-terminal region of ELAVL4 as the antigenic epitope, undergoing isoaspartylation, a spontaneous protein modification. This project explores the use of Protein Carboxyl Methyltransferase 1 (PCMT1) to identify isoaspartylation sites within ELAVL4 by exploiting PCMT1's ability to methylate isoAsp residues, enabling mass spectrometry detection. Recombinant PCMT1 was successfully expressed and purified under native and denaturing conditions, with preliminary enzyme activity assays underway. While initial results are inconclusive, optimizing detection ranges and storage stability will enable future mapping of isoAsp sites. This work aims to deepen our understanding of ELAVL4 isoaspartylation and its role in SCLC immunogenicity, with potential implications for biomarker development and therapeutic targeting.

Thanks to Shiori Fukutome, Diego Velarde and all the members of the Offringa lab for helping me with the experiments and suggestions within my time at USC. Thank you to the JumpStart Program for providing me with funding for this research opportunity.

63. Effects of Land Use on Mammalian Occupancy in Bolsa Chica Ecological Reserve

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Urbanization provides housing for growing human populations but leads to habitat loss and fragmentation, and a loss of biodiversity. While research has shown how different species adapt to coexisting with a growing human population, studies often overlook populations nestled within urban and wetland ecosystems. To help fill the gap in research, we are examining how different habitat types affect mesocarnivore occupancy and activity at Bolsa Chica Ecological Reserve. To monitor mammal occupancy, 10 motion-activated camera traps were deployed seasonally across different land cover types for at least 30-days (Jan,

Apr, July, Oct), with cameras installed across diverse habitat types. Species tagging for detections were done using the Urban Wildlife Information Network (UWIN) database while habitat data was extracted from a 10m land cover map via NOAA's Digital Coast data access viewer and ArcGIS. Coyotes and birds were the most frequently detected species across the reserve, highlighting their varied habitat use. The highest overall activity occurred at cameras in unconsolidated shore and near nesting sites, with 462 total detections across six species. Northern wetlands and riparian areas saw moderate activity, mainly from coyotes, birds, brush rabbits, and ground squirrels. Detections of striped skunks, fox squirrels, and raccoons were low in those same sites, suggesting lower species abundance than originally thought or that they are not triggering detections. Currently, the data is preliminary and only covers Year 1, as we still have four more seasons to deploy and four previous seasons that need to be analyzed.

This project is supported by the Bolsa Chica Land Trust.

64. Genetic Screening of Kinases and Phosphatases for Tumor Cell Dissemination in the Drosophila Midgut

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Dissemination is the key step for the initiation of metastasis, where tumor cells invade surrounding tissues and spread to distant sites. To study this process, we use the Drosophila midgut model expressing the oncogene Ras^{V12}, a mutation frequently observed in human cancers. In this system, Ras^{V12}-expressing intestinal stem and progenitor cells undergo morphological changes and extend invasive protrusions that breach the extracellular matrix (ECM) and the visceral muscle (VM), mimicking early events of tumor progression. Kinases and phosphatases are central regulators of signaling, functioning as molecular switches that turn pathways on and off through reversible phosphorylation. Because they control cell growth, adhesion, and migration, these enzymes are strong candidates for influencing tumor dissemination. To test this, we performed a candidate-based RNAi screen in the Drosophila midgut. Knockdown of Alp10, CycD, Pp4-19C, and Pez did not affect Ras^{V12}-driven dissemination. In contrast, knockdown of ADK3 and CycT significantly blocked dissemination after 3 days of induction, with tumors retained in the midgut. Moreover, we were able to quantify VM breaks, which serves as an indicator of the dissemination. Our study provides a platform to identify candidate kinases and phosphatases that regulate Ras^{V12}-driven dissemination in vivo, offering insight into the signaling pathways that drive tumor progression.

Special thanks to Vincente Abatay, Ginger Chiu, and Alan Vu for sharing their images.

65. Candidate based screen to identify potential upstream G protein-coupled receptors tasked with mediating Ca2+ influx in the disseminating *Drosophila Ras^{V12}* -expressing tumor cells

Izabella Thomas, Nathan Ocampo, Ginger Chu, and Jiae Lee, Ph.D.

Dissemination of cells from a tumor is the initial step in the process of metastasis; however, the molecular pathways involved in this process are still not well known. From previous results, it was found that intracellular calcium signaling induces the remodeling of E-cad/β-catenin subcellular localization, which is necessary for the invasiveness of *Ras^{V12}*-transformed cells in vivo. E-cad/ β-catenin disassembles at adherens junctions and assembles at invasive protrusions. This process is mediated by an intracellular influx of calcium. This calcium influx promotes the actin and cortactin-rich invadopodia-like protrusions that are essential for dissemination. Utilizing the *Ras^{V12}* tumor model in the Drosophila midgut, we sought to identify the upstream G protein-coupled receptors (GPCR) that regulate the calcium influx in *Ras^{V12}* expressing tumor cells.

This study is performed using a candidate-based approach of GPCRs, namely, mthl-5, mthl-15, Tre-1, and moody. Utilizing RNAi lines to specifically knock down the expression of these target genes in the RasV12 expressing tumor cells, we expect to observe the inhibition of dissemination within dissected midgut regions in GPCR RNAi flies that prevent calcium influx. Our initial screening has identified Tre-1 and mthl-15 as potential upstream GPCRs that regulate tumor cell dissemination. Moreover, we have confirmed the knock down of mthl-15 and tre-1 in the tumor cells shows the phenotype of disruption in dissemination. This study can illuminate an essential molecular aspect involved in initiating metastasis, which can provide potential targets in future

therapeutic applications.

Acknowledgements: Thank you to all the members of Dr. Lee's Drosophila Lab, NIH Bridges to Doctoral Grant for previous funding, and CNSM for access to microscopes and imaging software.

66. Investigating the Roles of Specific Cyclins in the Dissemination of *Ras^{V12}*-Driven Tumor Cells in *Drosophila melanogaster*

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Tumor cell dissemination is the initial step of cancer metastasis, which is responsible for the majority of cancer-related deaths. However, the molecular mechanisms that drive dissemination remain poorly understood. Through a candidate-based RNAi screening of kinases and phosphatases utilizing the *Drosophila melanogaster Ras^{V12}* intestinal tumor

model, we found that knockdown of specific cyclins blocks dissemination, whereas knockdown of others does not. Cyclins, which pair with Cyclin-dependent kinases (Cdk), are best known for their regulation of the cell cycle, however, they can also regulate other processes, such as transcription. Here, we investigate the role of Cyclin D and Cyclin T in tumor cell dissemination. Cyclin D is a cell cycle regulator that pairs with Cdk4 or Cdk6 to promote G1-S transition. Cyclin T, on the other hand, pairs with Cdk9, forming the positive transcription elongation factor (P-TEFb), which is essential for releasing paused RNA polymerase II, enabling productive transcription elongation. We found that Cyclin D knockdown had no observable effect on tumor cell dissemination, whereas knockdown of Cyclin T blocks tumor cell dissemination as well as resulting in a rounder and more stacked cellular morphology compared to the Ras^{V12} tumors. These results suggest that Cyclin T plays a distinct role in tumor cell dissemination from Cyclin D. Our findings reveal an underappreciated role of transcriptional cyclins in cancer metastasis and can provide potential targets for therapeutic strategies for preventing dissemination.

67. Generating Transgenic *Arabidopsis thaliana* Overexpressing *ERF54* by Floral Dip and Knocking out *ERF53* with CRISPR

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Leaf Senescence (LS) is a programmed degenerative process that recycles nutrients to other regions such as growing tissue. The molecular process begins with the dismantling of the chloroplasts, followed by the catabolism of proteins and lipids present in the cells. The *ERF54* gene is predicted to be a central regulatory hub in a gene regulatory network (GRN) for LS. However, testing this with an *erf54* single mutant showed an LS phenotype close to Wild-Type (WT). Looking at the sequence of *ERF54* revealed a potentially redundant paralog *ERF53*. To test whether *ERF54* is a central regulatory hub in the predicted GRN, we must generate transgenic *Arabidopsis thaliana* that are mutant alleles for both *ERF53* and *ERF54*. After these have been created we can move on to test how loss of function of these genes will affect LS phenotype and expression of other genes in the GRN. ERF54 overexpression (OE) lines are also needed to determine if gain of function for *ERF54* affect LS. Transgenic lines were made using the floral dip method with Agrobacterium tumefaciens to insert T-DNA randomly into the Arabidopsis thaliana genome. The OE vector was pEGG20 which includes a Ubiquitin-10 (Ubi10) constitutively promoted *ERF54* gene. Screening resulted in twenty-six potential hygromycin resistant transgenic *ERF54* overexpression lines. The most promising of which is the OX-7 line with a 150-fold-change for *ERF54* expression when compared to WT. The *ERF53* knockout lines were generated by floral dipping four pYUU-gRNA Addgene 159751 plasmid constructs that would direct CRISPR-Cas9 to target *ERF53* since we were unable to obtain T-DNA lines that disrupted *ERF53*. The CRISPR lines with active Cas9 are Yellow Fluorescent Protein positive (YFP +) and we used a YFP- progeny

because the active Cas9 would create new mutations in each generation. We isolated and sequenced DNA from these lines to establish which individuals had the *erf53* mutant indels resulting in frameshift mutations. Three lines have shown promising *erf53* null results, one mutant in the *erf54* background and two in the WT background. Sequencing for one of the mutants in the WT background was needed to ensure *ERF54* was not incorrectly targeted by CRISPR resulting in an off target *erf54* mutation during gene editing. Moving forward we hope to find at least one more stable OX line for ERF54 and then work with these *ERF54* and *ERF53* transgenic lines to test the predicted GRN. We aim to use WT, *erf53*, *erf54*, *erf53erf54*, and *ERF54*-OX as an opportunity to evaluate the role of ERF54 and its paralog, ERF53, in LS and other abiotic conditions.

68. Hydrogen Peroxide in the Apoplast during Leaf Senescence

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Leaf senescence (LS) is the development stage marked by chlorophyll degradation and nitrogen remobilization to developing tissues. The reactive oxygen species (ROS) superoxide anion ($O_2\bullet^-$) and hydrogen peroxide (H_2O_2) are signaling molecules synthesized in the apoplast. The production of ROS is essential for mounting defense responses to pathogens and mediating responses to environmental stresses, and for transitioning into LS. RbohD (AT5G47910) is a primary NADPH oxidase in the plasma membrane that produces extracellular $O_2\bullet^-$. This gets converted into the more stable H_2O_2 by superoxide dismutase (SOD). RbohD and RbohF (AT1G64060), are localized in the plasma membrane, and the mRNA for both genes is highly expressed in the leaves. PRX34 (AT3G49120, cell wall class III peroxidase) is soluble in the cell wall apoplast. RbohD, RbohF, and PRX34 contribute to the production of ROS in the apoplast. T-DNA insertion mutants in the ROS-producing genes *rbohD* (SALK_070610), *rbohF* (SALK_059888), and *prx34* (SALK_051769) are predicted to result in a reduction of apoplastic H_2O_2 and delayed LS. Higher order mutants *rbohDrbohF*, *rbohDprx34*, *rbohFprx34* and *rbohDrbohFprx34* are being produced to further explore the predicted delay in LS. To investigate the role of ROS during LS, DAB and NBT stains were used to observe the oxidative burst in wildtype leaves subjected to low light and high light stress. Younger leaves are expected to have no precipitate except at the wound site where the leaf was cut. Dark wound site staining was the basis for optimizing staining concentrations and post-vacuum incubation time. Chlorophyll levels were also measured. Superoxide anion accumulation detected with NBT blue precipitate staining is greater in photosynthetically active leaves, while hydrogen peroxide accumulation detected with brown precipitate stain by DAB is greater in yellowing senescent leaves. Lastly, the redox biosensor roGFP2-Orp1 will be used to observe and quantify H_2O_2 accumulation in the apoplast (Ingelfinger et al., 2025).

69. *ore1* Mutant Analysis: Delayed Leaf Senescence Under Variable Light Intensities

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Leaf senescence (LS) is a developmental process that recycles nitrogen and other nutrients to support new growth. ANAC092, also known as *ORESARA1* (*ORE1*, AT5G39610) is a NAC transcription factor and key regulator of leaf senescence; its overexpression accelerates senescence, while loss-of-function (LOF) *ore1* mutants display delayed senescence and a slight delay in flowering/bolting (Rauf et al., 2013). To study the role of *ore1* in bolting-induced LS, early events in LS, we examined in two independent T-DNA insertion alleles (SALK_090154 and SAIL_694_C04), which disrupt the *ORE1* gene at different positions. Using both alleles allows us to validate findings and minimize the possibility of allele-specific effects. Dark-induced leaf senescence (DILS) was used to confirm delayed LS phenotypes in both *ore1* alleles. Placing detached leaves in the dark is a starvation stress that induces LS. We next quantified bolting-induced LS (BILS) under low light and high light conditions. BILS is a transition to LS that is triggered by the vegetative to reproductive transition, which differs from stress-induced LS. High light adds a stress that should promote LS additively to bolting. Chlorophyll degradation was measured at time points T0 (day of bolting) and T12 (12 days after bolting), and senescence-associated genes (*NIT2* and *SGR1*) were quantified by real-time qPCR. Our initial experiments with one allele showed BILS was similar in WT and *ore1* at both light intensities. We are now testing the second allele. Our data suggest bolting-induced LS is not affected by ORE1 loss of function, even when accompanied by high light stress.

70. The Role of TET8-Associated and PEN1-associated EVs during Leaf Senescence in Arabidopsis thaliana

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Leaf senescence is the final stage of leaf development where cells of older leaves undergo metabolic and structural changes, leading to cell death. It is characterized by the yellowing of older leaves and the recycling of nutrients to be used for new leaf and floral development. Published work from our lab suggests that tetraspanin (TET) proteins are involved in leaf senescence (LS) and extracellular vesicles (EVs) may be regulators of LS in *Arabidopsis thaliana*. EVs are secreted in the leaf apoplast and have two described populations in *Arabidopsis*: TET8-associated EVs and PEN1-associated EVs. Zimmerman et al. 2024 showed that apoplast-localized TET8 along with TET3 may be positive regulators of leaf senescence. Therefore, I hypothesize that the overexpression (OE) of TET8 promotes LS.

pen1 and *pen1pen3* mutants showed early leaf senescence, suggesting that PEN1-associated EVs may be negative regulators of leaf senescence, opposite the role of TET8-associated EVs. Additionally, I hypothesize that early LS observed in *pen1pen3* is reversed by delayed leaf senescence observed in *tet3tet8*. A transgenic line containing a TET8 OE insert driven by a 35S constitutive promoter was obtained from UC Riverside and will be grown and screened to confirm TET8 OE at the mRNA and protein levels. A seed stock with a high percentage of *TET8* OE will be grown in soil for 6 weeks to compare LS to WT, LS phenotype will be measured by chlorophyll concentration, *NIT2* and *SGR1* expression transcripts, normalized to *ACT2*. *TET8* expression will be measured to verify *TET8* OE. In addition to the whole leaf, TET8 protein levels in the apoplast fluid will also be measured using Western blots to determine if high TET8 mRNA results in high TET8 apoplast protein levels. It is expected that *TET8* OE will display accelerated LS when compared to WT if apoplast TET8 shows high abundance. If this conclusion is drawn, it will strongly support the positive role of TET8-associated EVs in leaf senescence. *pen1pen3tet3tet8* quadruple mutants were identified through genotyping, and those seed stocks are being grown in soil for six weeks to undergo LS phenotype analysis, as previously described. The quadruple mutant will be compared to *pen1pen3*, *tet3tet8*, and WT. It is also expected that when both types of EVs are reduced, the *pen1pen3tet3tet8* mutants will display a LS phenotype closer to WT. This expected phenotype is consistent with EVs playing opposite roles in the regulation of leaf senescence.

71. TETRASPANINS and Their Role in Leaf Senescence

Luke Pickard, Sareh Darian, Ashley Ables, Thinh Phan, Connor Jenkins, Nhan Tran

TETRASPANINS (TET) proteins are a group of integral membrane proteins, with some being components of extracellular (EVs) in *Arabidopsis thaliana*. TET8-associated EVs are one of two EV populations described in *Arabidopsis*. Additionally, Zimmerman et al. 2024 demonstrated that *TET3* and *TET8* are positive regulators of LS. There are 20 *TET* genes in Arabidopsis and we are testing whether other TETs may also be associated with EVs or with LS. *TET7* and *TET18* were selected as the best candidates for this study since they were both present in published EV proteomes. *TET7* also shows high expression during leaf senescence and is closely related to *TET3* and *TET8*. *TET18* was selected for its high expression during leaf senescence. While breeding for the triple knockout mutant, *tet3tet7tet8* we have observed non-Mendelian segregation and have been unable to find individuals homozygous mutant for *TET3* and *TET7*. This suggests that tet3tet7 may be a lethal combination. One individual showed a late flowering and reduced fertility phenotype and was found to be homozygous *tet3tet8* and heterozygous *tet7*. Seeds from this plant are being collected and may contain a triple mutant. These seeds will also help clarify whether the late flowering and reduced fertility phenotypes are repeated in the next generation. Silique and germination studies will be performed to support lethality. Triple knockout *tet3tet8tet18* was successful and will be used for Nanoparticle Tracking Analysis (NTA) and

CSFE staining experiments. EV quantity from WT plants has been measured by NTA using a Nanosight NS300. Unexpectedly, the size of observed EVs has been variable. The results show one to two peaks, typically between 70nm and 250nm. Confocal microscopy with CSFE staining will be used to further investigate EV size and quantity.

72. Systemic Pesticide Transfer from Treated Corn to Wild Mustard Poses a Risk to Non-Target Herbivores

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Modern agriculture frequently uses chemical seed treatments combining fungicides and insecticides to mitigate yield losses caused by pests and pathogens. While effective, these compounds may unintentionally affect non-target organisms. New York State passed Senate Bill S8031, banning prophylactic neonicotinoid use starting in 2029. Diamides, considered less toxic to pollinators, are being considered as alternatives. Our project assesses whether diamide insecticides are safer by investigating whether non-crop plants can absorb pesticides from treated crops. We predict that a non-target plant, the wild mustard (*Rhamphospermum arvvense*), will take up pesticides from adjacent treated corn (*Zea mays*) plant, leading to a higher mortality of caterpillars feeding on wild mustard. We planted wild mustard next to corn seeds treated with the following seed coatings: untreated, fungicide only, fungicide with neonicotinoids, and fungicide with diamides. We calculated the relative growth rate of third star in star larvae of the fall army worm (*Spodoptera frugiperda larvae*) after feeding on wild mustard leaves for three days. In addition, we collected leaf tissue for LC-MS analysis to determine whether changes in caterpillar growth rate were associated with variations in secondary metabolite profiles. We found that caterpillars feeding on wild mustard leaves from plants growing next to the corn treated with diamide has lower relative growth rate than those fed on wild mustard grown next to the untreated control, fungicide-only, or neonicotinoid and fungicide treated corn. However, these results are preliminary, and further replication and chemical analysis are needed to confirm the observed trends. Therefore, any conclusions regarding ecological risk remain inconclusive at this stage. Nonetheless, our findings suggest that systemic insecticides may be taken up by non-target plants, potentially affecting associated herbivores and broader ecological interactions.

This project is supported by NSF award #1358843 and USDA award #2022-67037-36622.

73. Using a Probability Pavlovian-to-Instrumental Transfer Task to Investigate Withdrawal-Related Deficits in Rats

Isabella M. Monterosso, Jungyoung Choi, Connor A. Greene, Dr. Briac Halbout

Millions of Americans struggle with Opioid Use Disorder each year. During withdrawal, users often experience impairments in motivated and goal-directed behavior toward adaptive rewards, which can lead to continued drug use and relapse. These withdrawal-induced deficits pose a significant obstacle to recovery, making it necessary to better understand the neurobiological and behavioral mechanisms influencing these effects in order to develop more effective treatments. Within this broader scope, the present experiment aimed to develop a behavioral task to measure motivational and cognitive responses to reward-predictive cues, and to test whether those responses are altered by a disruption of endogenous opioid transmission in drug-naive rats. Specifically, we used a Probabilistic Pavlovian-to-Instrumental (Prob-PIT) task in male and female adult Sprague Dawley rats. This task assessed how two auditory cues, each predictive of reward delivery with a different probability, influenced reward-seeking actions, either promoting or inhibiting through their motivating and inhibitory control properties. At test, rats were injected with naloxone, a non-selective opioid receptor antagonist often used to reverse overdose, to model a withdrawal-relevant pharmacological state. We found evidence of differential, cue-specific influence on reward-seeking behavior, exposing the motivational and inhibitory control processes associated with these cues. This study will advance our understanding of how withdrawal impairs motivation to seek natural rewards, ultimately increasing vulnerability to relapse or drug-seeking behavior.

74. Investigating the Effects of Naloxone on Cue-Guided Decision Making in Drug-Naïve Rats

Cecelia R. Fanelli, Isabella Monterosso, Connor Greene and Dr. Briac Halbout

Opioid Use Disorder is a major problem in the United States. A major barrier to recovery is the withdrawal phase, which involves impairments in motivated and goal-directed behavior. These deficits are poorly understood, and a deeper investigation into their neurobiological and psychological underpinnings is needed. The following experiment serves as a pilot study for a larger project aimed at clarifying the mechanisms underlying withdrawal-associated disruptions in motivation and decision making. In this experiment we examine the relationship between opioid receptor activity and cue-guided behavior. Specifically, we investigate how naloxone—a non-selective opioid receptor antagonist commonly used to reverse overdoses—affects reward-based decision making in drug-naïve female and male Sprague Dawley rats. To do this, we employ two paradigms: outcome-specific Pavlovian-to-Instrumental Transfer (PIT) and outcome devaluation. PIT assesses how conditioned cues influence instrumental behavior, offering insight into the motivational processes that may be disrupted during opioid withdrawal. The devaluation paradigm distinguishes between goal-

directed actions and habitual responses by testing whether behavior remains sensitive to changes in reward value. By examining the effects of naloxone on motivated reward seeking, we aim to identify how blocking opioid receptors alters motivational processes. This project aims to uncover key links between motivated reward-seeking behavior and opioid receptor activity, providing a foundation for future research into the neurobiological mechanisms underlying opioid withdrawal and addiction.

75. Poster #75 reflects Abstract #39: DREAM Regulated Gene Expression in Wild Type and Knockout Zebrafish Larvae Exposed to TBBPA (presented twice)

76. Role of Cholesterol and Apolipoproteins in Maintaining Blood-Brain Barrier Integrity

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The blood–brain barrier (BBB) protects the brain by tightly regulating the movement of molecules between the blood and the central nervous system. It is formed by tight junctions (TJs) between brain endothelial cells, and cholesterol is essential for maintaining these junctions. Cholesterol can be provided from de novo synthesis or externally by high-density lipoproteins (HDL) or diet. Apolipoprotein E (apoE), a major component of HDL, exists in three isoforms (E2, E3, E4) and plays different roles in brain health, with apoE4 considered a risk factor for developing Alzheimer's disease (AD). We hypothesize that apoE facilitates cholesterol transport to TJs and that this process is isoform-dependent. To test this hypothesis, recombinant apoE2, apoE3, and apoE4 were expressed in a bacterial system, purified by affinity and, in select cases, by gel filtration chromatography. The purified proteins were reconstituted with commercially available phospholipids (POPC) and cholesterol to generate reconstituted HDL (rHDL). The rHDLs were applied to mouse brain endothelioma cells (bEnd 3) that have an upregulated level of low-density lipoprotein (LDL) receptors that are known to mediate receptor-mediated endocytosis. Cells treated with rHDL showed ~3-fold higher cholesterol levels compared to controls, confirming delivery of exogenous cholesterol. Western blot analyses revealed isoform-specific effects on TJ protein expression: claudin levels followed the trend apoE4 > apoE3 > apoE2, while occludin expression varied across two distinct bands, with apoE2 generally showing lower expression relative to apoE3 and apoE4. Ongoing experiments include tracking fluorescently labelled cholesterol to determine if it localizes to TJs, blocking endogenous cholesterol synthesis to test its contribution, and using Transwell assays to assess BBB permeability. By investigating how cholesterol reaches TJs and how apoE isoforms influence this process, our research aims to fill a critical knowledge gap in BBB maintenance. Understanding isoform-specific differences in cholesterol transport could provide insight into why apoE4 carriers are more susceptible to AD and may inform therapeutic strategies to strengthen BBB integrity.

This project is supported by the NIH NS# 140312 (VN), 2025 CSULB Student Summer Research Award and McAbee-Overstreet Graduate Research Scholarship (AA) and the 2025 CSULB MD Alum award (MH).

77. Conformational Analysis Of Reconstituted Discoidal And Spherical Hdl Bearing Apolipoprotein E3 And E4

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Apolipoprotein E (apoE) is a highly helical protein that plays a major role in cholesterol transport; it is found on high density lipoproteins (HDL)-like particles in the brain. In humans, it is polymorphic expressing two major isoforms, apoE3 and apoE4, the latter strongly linked with Alzheimer's disease. They differ in amino acid at position 112 (Cys and Arg, respectively). Our goal is to understand the conformational organization of apoE3 in reconstituted discoidal and spherical HDL, rHDL(d) and rHDL(s). We designed a series of single Cys variants of apoE3 with Cys located on various helices. Fluorescence emission spectra of pyrene-labeled lipid-free apoE3 variants revealed large excimer peak indicative of spatial proximity. A significant loss of excimer emission was noted in rHDL indicative of distance separation between two specified sites on neighboring apoE molecules. In parallel, rHDL with apoE3 single Cys variants were treated with bismaleimidohexane (BMH), which covalently links two proximal -SH moieties. Regardless of the location, all variants appeared as monomeric bands indicating that two neighboring apoE molecules were not aligned parallel on rHDL(d). Preliminary analyses of rHDL(s) reveal similar results, suggesting that despite the change of lipoprotein shape from discoidal to spherical geometries, two specified sites on neighboring apoE molecules are not proximal to each other. Taken together, our data indicate that 2 apoE molecules are aligned anti-parallel to each other on rHDL(d) and that the corresponding sites on 2 neighboring apoE are not juxtaposed to each other on rHDL(s). Our results offer valuable insights into the conformation of apoE3 and bear the potential to identify the mechanistic role behind the isoform-specific differences noted in brain HDL metabolism in neurodegenerative diseases.

This project is supported by grants from the National Institutes of Health #GM105561, #GMT34 149378, NSF LSAMP HRD-1826490 and the Richard D Green fellowship.

78. Effect of Oxidative Modification of Apolipoprotein E on the Cholesterol Status of Tight Junctions and the Integrity of the Blood Brain Barrier

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Alzheimer's disease (AD) affects over 55 million people worldwide. These rates could be expected to exceed 152 million by 2050. Apolipoprotein E (apoE) is a polymorphic protein whose primary role is cholesterol transport. Humans express three major isoforms including apoE2, apoE3, and apoE4, which differ at residues 112 and 158 by Cys or Arg substitutions. ApoE2 is associated with cardiovascular disease (CVD) but may grant protection against AD, apoE3 is largely cardioprotective, and apoE4 strongly correlates with CVD and AD risk. Notably, those homozygous for the *APOE* e4 allele are at a higher risk for developing AD compared to *APOE* e3 bearers. Previous studies in apoE knockout mice reveal apoE and age-related blood-brain barrier (BBB) leakage, implicating the importance of apoE in BBB maintenance. The BBB, formed by endothelial tight junctions (TJs), relies on cholesterol-rich domains and proteins such as ZO1, claudin, and occludin for barrier integrity. However, the mechanism by which cholesterol is supplied to TJs remains unclear. Previous studies on Alzheimer's patients revealed oxidatively modified proteins (modified by 4-Hydroxynonenal (HNE), a reactive oxygen species) present within the blood vessels of the brain. HNE is a product of lipid peroxidation and oxidative stress, which are hallmarks of aging; it modifies proteins and alters their function.

We hypothesize that oxidative modification of apoE by HNE impairs its capacity to deliver cholesterol to endothelial TJs, thereby weakening BBB integrity. To test this hypothesis, recombinant human apoE2, apoE3, and apoE4 bearing a hexa-His tag at the N-terminal end were expressed in *E. coli*, purified via Ni-affinity and gel filtration chromatography, and treated with HNE. Western blotting using anti-HNE antibody confirmed oxidative modification, although faint bands in ethanol-treated controls suggest possible air oxidation requiring further investigation. rHDL particles were generated by reconstituting apoE isoforms with cholesterol and POPC. Fractions were characterized by ultracentrifugation and BCA assay, with fractions pooled for subsequent cell studies. Preparations of rHDL containing unmodified and HNE-modified apoE isoforms are being optimized prior to application to bEnd.3 endothelial cells, where permeability assays, TJ protein quantification (ZO1, claudin, occludin normalized to β-actin), and TJ cholesterol content will be measured. We anticipate that HNE-modified apoE will be defective in cholesterol delivery to the TJs in bEnd.3 endothelial cells and there may be apoE isoform specific differences in the status of TJ markers between the HNE-modified apoE. Data from our studies are expected to yield further insights into the role of apoE in modulating the integrity of BBB, which will improve our ability to devise therapeutic intervention strategies to combat oxidative stress related events in AD and aging.

79. Epigenetics of Environmental Risk Factors: Evaluating the Earthworm as a Model for Sporadic Neurological Disorders

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Environmental risk factors such as diet and pollutants can influence DNA methylation, affecting protein production and cellular function. DNA methylation is an epigenetic mechanism involving the transfer of a methyl group onto the C5 position of cytosine to form 5-methylcytosine, and it relies on DNA methyl transferases such as DNMT1, 2, and 3 [1]. In the central nervous system, DNA methylation shows a complex repertoire of epigenetic gene regulation that could offer therapeutic targets for neurological disorders. In particular, neuronal activity can modulate the pattern of DNA methylation in response to physiological and environmental stimuli [2]. As such, we set out to establish a model system that would help clarify the functional links between epigenetic changes due to sporadic triggers of neurological disorders and their molecular and cellular pathways. Research using earthworms such as *Lumbricus terrestris* shows that their epigenome displays a high level of DNA methylation patterns that can change in response to environmental pollutants and confirms the presence and function of DNMTs [3]. Earthworms have also been widely used to evaluate histopathological and molecular changes induced by environmental toxicants [4]. They possess a simple nervous system consisting of a medial ventral nerve cord that extends along the length of the body and terminates at two cerebral ganglia that make up a primitive brain. Our ongoing work has demonstrated neural spike recording to quantify spike conduction velocities along the length of the nerve cord as a measure of cellular function. Concurrently, we are performing optically guided worm dissections to isolate the nerve cord and assay epigenetic responses by assessing levels of DNMT1, 2, and 3, combined with histopathology analysis to evaluate whether persistent exposure to bisphenol A (BPA) and β-sitosterol-β-d-glucoside (BSSG) can lead to neurotoxicity. Given that BPA and BSSG are ubiquitous environmental toxins and known triggers of neurodegeneration [5], our systematic approach would establish a viable model to further understand sporadic pathways of complex neurological diseases.

References: [1] PMID: 22781841; [2] PMID: 22781841; [3] PMID: 35878310; [4] PMID: 35202286; [5] PMID: 30323119

Acknowledgments: We are grateful for the Chemistry & Biochemistry Department Dr. Roger A. Acey Endowed Summer Research Assistantship to AT and the CSULB start-up support to SV. We thank Mary Nguyen for her contributions.

80. Cognitive Dysfunction Parallels Astrocyte Reactivity in the SOD1^{G93A} Mouse Model for Amyotrophic Lateral Sclerosis

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Amyotrophic lateral sclerosis (ALS) is a neurodegenerative disease affecting ~1/100,000 Americans, with presently no cure. Motor neurons (MNs) which control the skeletal muscles progressively degenerate in ALS, leading to debilitating paralysis and fatality. The transgenic SOD1^{G93A} mouse with a point mutation in the Super-Oxide-Dimutase-1 gene has been a crucial model of ALS, advancing the mechanistic knowledge that not all MNs are equally vulnerable to degeneration [1]. Supporting this, we previously showed postnatal abnormalities in the membrane excitability of ALS-vulnerable jaw controlling MNs (JMNs) and their proprioceptive sensory neurons, while the ALS-resilient extraocular (EO) MNs remained unchanged [2, 3]. Additionally, using transcriptomic analysis, we identified brain-wide changes in glial cells including astrocytes indicating non-cell autonomous contributors to disease development [4]. Based on this evidence, we hypothesize that astrocyte molecular signatures differentiate MN vulnerability markers in ALS. To test this, we are evaluating the longitudinal expression of astrocyte reactivity markers nearby JMNs and EO MNs that would further coincide with detectable MN degeneration and early behavioral dysfunctions. Our current results indicate upregulated glial fibrillary acidic protein (GFAP) expression, an astrocyte reactivity marker, in the vulnerable brainstem jaw motor nuclei, and its absence in the resilient EO motor nuclei. Secondly, behavioral data indicates an unexpected cognitive dysfunction at the early stage of disease development, when motor symptoms are not yet significant. Ongoing work is examining whether astrocyte reactivity occurs in close proximity to degenerating MNs, at early stages, further supporting astrocyte-specificity driving MN vulnerability in ALS.

References: [1] PMID: 20367447; [2] PMID: 25589764; [3] PMID: 31530644; [4] PMID: 32360664

Acknowledgments: We are grateful for the UCLA Summer Research Scholarships to MM and BM, CSULB BUILD Program with OURS to JR, and CSULB startup support to SV. We thank Dr. Martina Wiedau at UCLA for her support and contributions.

81. A Novel Multiscale Molecular Network Model of Functional Pleiotropy Elucidates Neuroimmune Crosstalk

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Functional pleiotropy refers to a broad spectrum of molecular and cellular effects caused by a single gene or molecule influencing distinct aspects of the organism's physiology [1]. For example, a single cytokine such as TNF-α can influence both immune responses and neural excitability and neurotoxic pathway in the brain [2]. However, information on which cells secrete which proteins, and how these proteins influence other cell types, is often dispersed across individual studies. In this work, we curated data from primary research articles (PRAs) published between 2002 to 2024. We extracted information on source cell type (e.g., neurons), secreted biomolecules (e.g., neurotransmitters), target cell type (e.g., microglia), and resulting functional effects (e.g., increased cytokine signaling) from each PRA. We then developed a step-by-step workflow to build a network graph model to visualize the consolidated molecular interactions from 70 PRAs. Using neuron–microglia signaling as an example, we show how interactions between neuroinflammatory and neural excitability functions are in a feedback loop of molecular interactions. We introduce a Functional Interaction Score (FIS) that captures the magnitude and direction of literature evidence. Our systematic pipeline of activities can be widely adopted to develop multiscale network models of molecular interactions, known as Functional Interactomes Across Brain Scales (FIABS), which integrate diverse findings into unified networks. The case of neuroimmune crosstalk demonstrated in this study identifies significant neuroimmune modulation of neural excitability by microglial cytokines, which alter the expression and function of intrinsic and synaptic ion channels in neurons. This framework provides a new tool to investigate nervous system function and dysfunction, with potential applications in understanding disease mechanisms.

References: [1] PMID: 36353359; [2] PMID: 26112872

Acknowledgment: This work was partly supported by the CNSM Leslie K.wynston Summer Research Assistantship to MZY and by the CSULB startup funds to SV. We thank Jakob von Morgenland, Jasmine Motawakel, RoxanneKnorr, and Artin Allahverdhian for their contributions.

82. Functional Impact of the L233S Mutation in Apoliprotein A-I Lipid Sensing

Shreya Thakuri, Paul M.M. Weers

Apolipoprotein A-I (apoA-I) is the primary protein found on high-density lipoproteins (HDL) and plays a critical role in reverse cholesterol transport. Reverse cholesterol transport is a process which is essential for the removal of excess cholesterol from peripheral tissues and

reducing the risk of cardiovascular disease. When this process is disrupted or is inefficient, it can lead to the development of atherosclerotic plaques which can result in a heart attacks and strokes. Higher levels of HDL cholesterol are linked to lower cardiovascular risk, while low HDL levels or dysfunctional HDL leads to an impaired cholesterol clearance causing increased cardiovascular risk. ApoA-I needs to bind to phospholipids to form discoidal HDL (lipidation) and does this through the C-terminal α-helices (H8-H10). We Hypothesize that Helix 10 (H10), serves as the hydrophobic sensor due to its amphipathic and hydrophobic nature and exposed location in the protein. This study focuses on gaining a better understanding on the role of H10 in lipid sensing by introducing a targeted point mutation. By site-directed mutagenesis the L233S mutation was introduced to disrupt the non-polar face of H10. This could lead to decreased ability to form discoidal HDL. After expression and purification of L233S-apoA-I, lipid binding will be analyzed by solubilization of phospholipid vesicles. This should lead to a better understanding of the role of the C-terminal domain in the function of this critical plasma protein.

Acknowledgements: I would like to thank all my lab members for their valuable assistance, mentorship, and support throughout this research project. I am also grateful for Dr. Paul Weers for his guidance and encouragement. Lastly, I want to thank the CSULB M.D. Alumni Program for funding this research opportunity.

83. Evaluating Fmoc-amino esters as selective butyrylcholinesterase inhibitors as possible preventative therapeutic for Alzheimer's disease

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Alzheimer's disease (AD) is a progressive neurodegenerative disorder associated with decreased levels of the neurotransmitter acetylcholine, leading to cognitive decline and memory loss. While *acetylcholinesterase* (AChE) and b*utyrylcholinesterase* (BChE) both degrade acetylcholine, studies suggest that BChE activity increases in AD patients, making it a promising therapeutic target. We previously identified *9-fluorenylmethoxycarbonyl* (Fmoc)-amino acid derivatives as selective BChE inhibitors. Nevertheless, the negatively charged carboxylate in the original series may be a limitation, as the enzyme catalyzes the hydrolysis of a cationic acetylcholine substrate. To this model, this study investigated the selective inhibition of BChE using esterified analogs versus the carboxylate forms. Fmoc-leucine, -valine, -methionine, -tyrosine, and -phenylalanine were synthesized or obtained in both ester and carboxylate forms and tested using a relative enzyme activity assay measured via spectrophotometry. A general trend emerged showing that esterified compounds demonstrated greater inhibition than their carboxylate counterparts. Of the Fmoc-ester compounds used, methionine accounted for the most inhibition at an average of 20% enzyme activity. These results suggest that the ester functional group enhances the compound's interaction with BChE and supports the model that the negative charge is

unfavorable for inhibition. This study highlights the potential of Fmoc-ester derivatives as selective BChE inhibitors and supports further investigation into their active site interactions, structural optimization, and the potential reversibility of inhibiton. The findings may contribute to the development of more effective treatments aimed at slowing the progression of Alzheimer's disease by preserving acetylcholine levels through targeted BChE inhibition.

Acknowledgements: This work is supported by the CSULB BUILD Program in the Office of Undergraduate Research Services.

84. Poster #84 reflects Abstract #48: Single-cell detection of cryptic viral and parasite infections in marine diatoms (presented twice)

85. Investigating Enantioselectivity of Bisoxazoline Ligands in Fe-Catalyzed Kumada Cross-Couplings

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Transition metal–catalyzed cross-coupling reactions are essential in pharmaceutical and industry synthesis. While nickel catalysts are well-studied, their toxicity and cost limit large-scale use. Iron offers a cheaper, less toxic alternative, but its potential in asymmetric transformations remains underexplored. Our laboratory is investigating iron-catalyzed Kumada cross-couplings with bisoxazoline (BOX) ligands to evaluate their ability to promote enantioselectivity. Current efforts focus on optimizing conditions such as temperature, scale, and flow rate to improve enantiomeric excess (ee). Preliminary results highlight the strong influence of ligand structure and reaction parameters on selectivity. This work aims to expand the understanding of iron catalysis, establishing a foundation for sustainable cross-coupling methodologies with broad applications in both pharmaceuticals and industrial chemistry.

This project is supported by the American Chemical Society Petroleum Research Grant 69209-UNI1.

86. Optimization of Ni-Catalyzed Reductive Cross-Coupling with NHP Esters and Vinyl Bromides Reaction Conditions

Joe Rocha, under the direction of Dr. Julie L. H. Wahlman, Chemistry and Biochemistry

Metal-catalyzed cross-coupling reactions are crucial techniques used in the formation of carbon-carbon bonds in complex bioactive molecules. In nature, compounds containing a dihydrobenzofuran core possess bioactive properties that display anti-inflammatory,

anticancer, and antibiotic effects. In our laboratory, we are interested in using a synthetic approach utilizing Nickel-Catalyzed Reductive Cross-Coupling reactions to create dihydrobenzofuran derivatives. This further enables late-stage diversification, exploring the bioactive efficacy when manipulating variables such as functional groups and stereoisomers. Reactions involving a metal catalyst often depend on ligands, as they are typically unstable on their own. Ligands ultimately dictate factors such as product yield and stereoselectivity by regulating catalytic stability, reactivity, and desired reaction pathways. As ligands are crucial towards the synthetic production of dihydrobenzofurans, we are ultimately interested in investigating box-ligands to observe how they affect the product enantioselectivity between vinyl bromides and NHP Esters when cross-coupled with a nickel catalyst. To support these future ligand screening investigations, optimization of the reaction was done to improve product yield through experimental factors such as temperature, solvent ratio, and Lewis acid additives. Through optimization, consistent baseline reaction conditions were determined. With greater reliability, differences in yield and enantioselectivity found in future ligand screenings can now be confidently attributed to the experimental box ligands. Reaction screening efforts and their results will be discussed.

Project funded by faculty start-up funds provided by CSULB College of Natural Sciences and Mathematics.

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87. Photoredox-Enabled C–H Alkylation via Catalytic Samarium(II)-Mediated Metalloradical–Radical Coupling

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Direct formation of carbon–carbon bonds from unactivated C–H sites is a long-standing challenge in synthesis, particularly for dissonant bond disconnections and when conventional organometallic reagents are impractical. Samarium(II) reagents are attractive for such transformations due to their low cost, chemoselective single-electron reactivity, and radical compatibility, but stoichiometric use is hindered by low concentration, poor scalability, and incompatibility with asymmetric catalysis. To address these limitations, we

developed a ternary catalytic system that integrates visible-light photoredox activation, samarium(II)-mediated metalloradical–radical coupling, and tertiary amine hydrogen atom transfer (HAT) catalysis to achieve C–H alkylation of ketones. Using benzylacetone and tetrahydrofuran (THF) as model substrates, initial reaction discovery gave 8% yield. Systematic optimization under rigorously anhydrous conditions identified key parameters influencing reactivity, including photocatalyst identity, HAT catalyst structure and loading, substrate stoichiometry, and additive compatibility. Quinuclidinyl benzenesulfonate at 1.0 equivalent emerged as the optimal HAT mediator, balancing radical generation while minimizing Sm deactivation by over-coordination. The best conditions was $Sm(OTf)_3$ (1.0 equiv), Ir-based photocatalyst (2 mol%), quinuclidinyl benzenesulfonate (1.0 equiv), and THF (5 equiv) in acetonitrile under 450 nm irradiation for 2 hours, which resulted in the C–H alkylation products up to 45% yield. Across all studies, coordination effects from solvents, additives, and HAT catalysts were identified as critical determinants of samarium solubility and turnover. These results establish proof-of-concept for catalytic Sm(II)-mediated metalloradical–radical C–C bond formation and highlight strategies to improve reactivity and radical capture. Ongoing efforts focus on designing less basic HAT mediators and expanding scope to dioxolanes and aldehydes, ultimately laying the foundation for chiral and asymmetric studies. Catalytic samarium(II) chemistry offers a promising pathway for carbon–carbon bond formation under mild and sustainable conditions.

This project is supported and funded in part by the Resnick Sustainability Institute – Caltech WAVE Fellows program.

88. Role of AnnexinA2 in HIV-1 Nef-Dependent Infectivity

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Human Immunodeficiency Virus Type-1 (HIV-1) encodes accessory proteins (*vif, vpu, vpr, and nef*) that modulate host immunity. Nef (negative factor) is a peripheral membrane protein expressed early in the HIV-1 replication cycle. Nef supports several pro-viral functions, including downregulation of cell surface receptors, modulation of T cell receptor signaling, and enhancement of virion infectivity. However, the mechanisms by which Nef increases infectivity remain unclear.

Using virion proteomics, we identified host proteins modulated by Nef. One of these proteins was AnnexinA2 (AnxA2), a membrane-associated calcium- and lipid- binding protein involved in inflammation, cell signaling, and membrane organization. Diverse viruses hijack AnxA2 to enhance viral replication. Using Bi-Molecular Fluorescence Complementation (BiFC), we found that Nef physically interacts with AnxA2. Therefore, we hypothesized that Nef requires AnxA2 to enhance viral infectivity and replication.

Our preliminary results show AnxA2 overexpression increases HIV-1 infectivity. To further investigate this relationship, we generated a CRISPR/Cas9-derived AnxA2 knockout HeLa cell line. These cells will be transfected with plasmids encoding either wild-type or Nef-deficient HIV-1, and viruses produced will be assessed for infectivity. We also aim to knockout AnxA2 in primary CD4+ T cells using CRISPR Ribonucleoproteins (RNPs) and measure viral infectivity to enhance the physiological relevance of our findings. Together, this work will clarify how Nef enhances viral infectivity and may inform strategies to develop novel Nef inhibitors.

89. Determining the role of AnnexinA2 in SARS-CoV-2 replication

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Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2) has caused more than 7 million deaths worldwide. Although vaccines are available, few antivirals exist, underscoring the need to understand better host factors required for viral replication. Annexin A2 (AnxA2) is a phospholipid-binding protein implicated in multiple viral infections, though its importance in SARS-CoV-2 replication remains unclear.

Our preliminary data suggest that SARS-CoV-2 Spike interacts with AnxA2. Based on this, we hypothesized that Annexin A2 is required for the entry and/or assembly of SARS-CoV-2. We identified two siRNAs that efficiently knocked down Annexin A2 in a human lung cancer cell line (A549) and confirmed ACE2 and TMPRSS2 surface expression by flow cytometry. We generated replication-incompetent lentiviruses pseudotyped with different envelope glycoproteins: vesicular stomatitis virus glycoprotein (VSVG), which served as a control, or SARS-CoV-2 Spike. Preliminary results showed low infection efficiency of DHIVGFP-Spike compared to robust infection with DHIVGFP-VSVG, suggesting further optimization is required.

To determine whether Annexin A2 contributes to viral assembly, A549 wild-type or AnxA2 knockdown cells will be transfected with plasmids encoding SARS-CoV-2 structural proteins (Envelope, Membrane, Nucleocapsid, and Spike) together with a luciferase reporter. The resulting virus-like particles (VLP) will be collected and used to infect target cells, and luciferase activity will be measured as a readout of infectivity. Reduced infectivity of VLPs produced in Anxa2-deficient cells would indicate a role in viral assembly. To evaluate the role of AnxA2 in viral entry, A549 wild-type or AnxA2 knockdown cells will be infected with DHIVGFP-VSVG and DHIVGFP-Spike. Infectivity will be quantified as the percentage of GFP-positive cells by flow cytometry. Lower infection by Spike pseudoviruses in AnxA2 knockdown cells would implicate AnxA2 as a host factor required for viral entry. Together, these studies will define whether Annexin A2 contributes supports SARS-CoV-2 assembly or entry, providing insight into its potential as an antiviral target.

Acknowledgment: I thank Dr. Peter Ramirez and the Ramirez lab for their guidance in this project. This work was supported by NIHR16AI184450 awarded to PWR.

90. A SARS-CoV-2 Virus Like Particle (SC2-VLP)-Assay to Evaluate Heparan Sulfate-Like Compounds for use as broad-spectrum antivirals

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Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2), the causative agent of the COVID-19 pandemic, is an enveloped, *betacoronavirus* with a positive sense single stranded RNA (+ssRNA) genome. It shares approximately 80% genetic similarity to SARS-CoV and primarily infects epithelial cells of the upper and lower respiratory tract. While antivirals such as Paxlovid and Remdesivir are available, none target the earliest stages of infection that drive SARS-CoV-2 transmission. Heparan Sulfate Proteoglycans (HSPGs) are cell surface proteins containing negatively charged (HS) polysaccharides, facilitating attachment of diverse viruses such as HIV-1 and SARS-CoV-2. Synthetic HSPG mimics known as Sulfoglycodendrimers (SGDs) block HIV-1 binding, but their effects on SARS-CoV-2 remain unclear.

To address this, we used a SARS-CoV-2 Virus Like Particle (SC2-VLP) system to test the impact of a synthetic polysaccharide analog, dextran sulfate (DS), on SARS-CoV-2 entry. SC2-VLPs contain all four structural proteins (Spike, Envelope, Matrix, and Nucleocapsid) and package a luciferase reporter, allowing quantitative measurement of entry and assembly without requiring infectious virus. We hypothesized that dextran sulfate would reduce SC2-VLP infectivity by competing with HSPG-mediated attachment. Target cells expressing ACE2 and TMPRSS2 were pre-incubated with varying concentrations of DS before or after SC2-VLP exposure. A luciferase assay assessed infectivity, while metabolic assays assessed cytotoxicity.

Pre-treatment with sub-micromolar DS (10-100nM) reduced SC2-VLP infectivity by 60-80%, with minimal effects on viability (>85%). Adding DS after infection yielded variable results. These findings establish SC2-VLPs as a robust model to study viral entry inhibitors of SARS-CoV-2 and suggest sulfated polysaccharides may serve as promising SARS-CoV-2 antivirals. Future work will evaluate authentic SGDs, including sulfated colominic acid, sialic acid, and maltotriose.

91. The Molecular Residue T221 Unlocking CDK5-p25/35 Control of Cell Activity

Iris Marquez (no longer presenting)

92. Salicylate Treatment Activates IRE1a Signaling in Colorectal Cancer

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Acetylsalicylic acid, commonly known as aspirin, is an over the counter, non-steroidal, anti-inflammatory drug that has been linked to reduced occurrence of colorectal cancer, but the cellular mechanism(s) for its chemoprevention remains elusive. Cancer cells have altered metabolic demands due to their hypoxic environment and rapid proliferation rate that can induce proteotoxic stress. During this stress, cells activate an evolutionarily conserved signaling pathway known as Unfolded Protein Response (UPR) which has been linked to the acquisition of malignancy and chemoresistance in cancer cells. The UPR is governed by three transmembrane protein sensors known as protein kinase R-like endoplasmic reticulum kinase, activating transcription factor-6, and inositol requiring enzyme-1a (IRE1a). With a central hypothesis that aspirin mediated chemoprevention involves regulation of the UPR, the goal of this work is to test the effect of aspirin treatment on UPR in two colorectal cancer cell lines (DLD1 and HCT116). DLD1 and HCT116 colorectal cancer cells were treated with sodium salicylate, a metabolic derivative of aspirin, and the activation of the UPR sensor IRE1a was analyzed via western blotting. The downstream splicing of the X-box binding protein-1 (XBP1) mRNA by activated IRE1a was confirmed via reverse transcription polymerase chain reaction as well as by western blotting to directly assess XBP1 expression. Together, our results confirm that IRE1a signaling is activated in salicylate treated colorectal cancer cell lines. Our future experiments will focus on establishing a Drosophila model to study the role of IRE1a in salicylate-mediated chemoprevention.

Acknowledgements: We would like to thank Dr. Judy Brusslan and Kelsey Ligos for their help in developing an RNA isolation protocol. This project is supported by an NIH grant to Dr. Bhandari (Grant # SC3GM139707). Amber Peek is supported by the Leslie K. Wynston Summer Research Assistantship and NIH Bridges to the Doctorate program (Grant #5T32GM138075-05). The content is solely the responsibility of the authors and does not necessarily represent the official views of the NIH.

93. Employing a Click-Chemistry-Based Approach to Investigate Protein Targets of Aspirin in Colonic Organoids

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Long-term daily regimens of aspirin have been demonstrated to reduce the risk of colorectal cancer – the second most common cause of cancer deaths in the U.S. However, the

biochemical pathways and mechanisms behind this prophylactic effect remain elusive and current investigative work is limited to cell culture-based models. Previous groups have employed an aspirin-based small molecule probe to enrich and identify aspirin-binding proteins in live cells. We sought to expand this technique for use in the more complex and physiologically relevant model of patient-derived organoids. Using an alkyl-modified probe, we subjected human colonic organoids to a click-chemistry based pull-down followed by label-free quantification mass spectrometry analysis. From this work, we identified 78 high-significance protein targets that had likewise been previously reported from cell culture-based studies. Gene ontology analysis of these 78 targets revealed significant enrichment of proteins involved in protein folding and translation as well as those capable of binding RNA. Preliminary work is underway to conduct directed studies in cell culture models to probe the impact of aspirin treatment on these targets and their associated cellular pathways.

Acknowledgements: The authors gratefully acknowledge Dr. Julie Wahlman and Zachary Hill for their assistance in synthesizing the aspirin-based probe as well as the generous support of CSULB Graduate Equity Fellowship and the Leslie Wynston Summer Research Assistantship to LM. This project is supported by an NIH grant to Dr. Bhandari (Grant # SC3GM139707). The content is solely the responsibility of the authors and does not necessarily represent the official views of the NIH.

94. Targeted Delivery of Salt Inducible Kinase Inhibitors with Zoledronate Coated ZIF-8

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Metal-organic frameworks (MOFs) are highly porous, crystalline materials composed of metal ions connected by organic linkers, offering remarkable potential for drug delivery applications due to their large surface area, low toxicity, and biocompatibility. Zeolitic Imidazolate Framework-8 (ZIF-8) stands out as a well-regarded MOF known for its solvo and thermal stability, easily synthesized at room temperature through the self-assembly of zinc ions and 2-methylimidazole linkers. We explored the drug loading capacity of ZIF-8 for salt-inducible kinase (SIK) inhibitors — molecules with potential therapeutic relevance in managing osteoporosis, a condition characterized by weakened bones and increased fracture risk. To enhance the framework's performance and biocompatibility, we applied a bisphosphonate coating, specifically using zoledronate, known for its strong affinity to bone tissue and ability to inhibit bone resorption. 2-tert-Butyl-910-di(naphth-2-yl)anthracene (Proxy Molecule 1) and 5-Amino-1,3-dimethylpyrimido[4,5-d]pyrimidine-2,4(1H,3H)-dione (Proxy Molecule 2) were used to mimic SIK inhibitor 1 (HG-9-91-01) and SIK Inhibitor 2 (YKL-05-099), respectively. Thermogravimetric analysis (TGA) and Infrared spectroscopy were used to confirm the encapsulation of the proxy molecules and to ensure zoledronate coating on the surface of ZIF-8. High-Performance Liquid Chromatography (HPLC) was performed to

measure the concentration of proxy being eluted from ZIF-8 when in an experimental environment mimicking blood pH, temperature, and movement. Additionally, X-ray diffraction (XRD) was used to confirm ZIF-8 structure and chemical composition remained. This research aims to optimize MOF-based drug delivery systems for bone-targeted therapies.

Acknowledgements: Research reported in this project was supported by the National Institute of General Medical Sciences of the National Institutes of Health under Award Number SC3GM136590. The content is solely the responsibility of the authors and does not necessarily represent the official views of the National Institutes of Health. Keck Undergraduate Research Experiences (KURE) Incubator for financial support to this project.

95. Phasefield Modeling of Bacterial Colony Formation with Multigrid Method and GPU Acceleration

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Bacterial crossfeeding exhibits interesting pattern formation. For example, a colony composed of a producer strain which consumes lactose and produces galactose for the consumer strain may display flower pattern. In order to understand this kind of pattern formation, we propose a 3D phasefield model to simulate bacterial growth and colony formation on an agar dish in an experimental setting. The model is composed of some Partial Differential Equations on the phasefield describing the colony region, the densities of two bacterial strains, the underlying nutrients, the pressure, and the velocity field. In order to solve the system of equations efficiently and accurately, we use Multigrid Method and Semi-Lagrangian Method, with GPU acceleration. To validate our model, we simulate a colony with one single species. Our results show that colony height has a much stronger dependance on the nutrient supply than the radius, consistent with [Warren et. al, eLife, 2019]. Our two species crossfeeding simulation generates nice radially symmetric patterns. But we still haven't observed flower patterns yet, which might be due to the lack of symmetry breaking mechanism in our model. As a next step, we plan to add some randomness into the model to break the symmetry.

This project is supported in part by NIH grant #R35GM152133 and CSULB Summer Student Research Assistantship.

96. Multilingual Mathematical Proof-Writing Sessions

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Multilingual mathematics majors rarely get an opportunity to attempt mathematics in languages other than English. The study examined whether conducting multilingual proof-writing groups had any effect on bilingual mathematics students. The participants consisted of eight bilingual mathematics majors who speak English and Spanish. They participated in proof-writing sessions weekly, where participants used proofs from their homework and wrote them in English and Spanish. Five of the participants completed a survey after the conclusion of the proof-writing sessions and four of those participants took part in follow-up interviews nine months after the proof-writing sessions. To analyze the data, we wrote and compared initial notes and then completed a round of coding to assess similarities and important findings. The codes were then further analyzed to fully evaluate the impacts of the proof-writing sessions. Based on the analysis, participants reported several positive effects from the proof-writing sessions. These effects included improving their proof-writing skills, bonding with other students who had similar backgrounds, considering using Spanish to complete mathematics in the future and feeling a greater sense of identity and belonging in their academic environment. These discoveries highlight that there could be benefits to allowing students to complete mathematics in multiple languages and could provide a foundation for future studies.

97. Investigating the Effect of 2,6-diarylidene cycloalkanones on Akt and Integrated Stress Response

Oshiana Schenkelberg, Deepali Bhandari, Ph.D.

Protein kinase B, also known as Akt, is a serine/threonine kinase that plays pivotal roles in many important cellular processes including growth and proliferation, survival, migration, and metabolism. Dysregulation of Akt has been associated with several human diseases including cancer making it a therapeutic target for cancer treatment with many small molecule inhibitors currently in clinical trials. Our laboratory utilizes commercially available Akt inhibitors to study the role of Akt in cancer cell survival. This work focuses on one such inhibitor, SC66 ((2E,6E)-2,6-Bis(4-pyridinylmethylene)-cyclohexanone). SC66 was first tested and published as an Akt inhibitor, with the study suggesting that treatment with SC66 inhibited Akt activity and resulted in reduced cell proliferation and apoptosis. However, our lab discovered that treatment with SC66 led to activation of Akt and the integrated stress response (ISR) in multiple cell lines. This work aims to understand the mechanism by which SC66 activates Akt and ISR, and to test whether SC66-induced Akt activation and ISR activation are interdependent. Using a cervical cancer cell line called HeLa, and a colorectal adenocarcinoma cell line called DLD1, we analyzed three structural analogs of SC66 to

compare their outcome with that of SC66 treatment on HeLa and DLD1. Lysates of the cells treated with either the analogs or SC66 were analyzed by western blotting. Using specific antibodies that detect the activated form of Akt and ISR effector molecules, we determined that a substitution at the 4-position of the cyclohexanone with a phenyl group maintained the Akt and ISR activation, but replacing the carbonyl group with a hydroxyl group or replacing the cyclohexanone with a cyclopentanone abolished SC66-dependent Akt and ISR activation. For our second goal, we co-treated cells with SC66 and a known ISR inhibitor (ISRIB) to test if Akt activation is dependent on ISR activation. Lysates of the cells treated with SC66, ISRIB, or both were analyzed by western blotting. Our results suggest that Akt activation is independent of ISR activation in SC66-treated cells. The future experiments will focus on determining if SC66-mediated ISR activation is dependent on Akt activation. The findings of this study suggest that SC66 and its analogs cause cell death by triggering cellular stress and not due to Akt inactivation as previously proposed, thus shedding light on the correct mechanism of action of SC66.

This project is supported in part by the National Institutes of General Medical Science Grant SC3GM139707 and by the National Institutes of General Medical Science T34 Grant GM149378-01. We thank Dr. Nicholas Salzamenda at CSU Fullerton for synthesizing and providing us with the DC compounds.

98. Pore-Space-Partitioned Metal Organic Framework Synthesized from Chiral Camphoric Acid

Jude Saxton, Dr. Xianhui Bu

Metal-organic frameworks (MOFs) are a diverse class of crystalline materials characterized by high porosity, geometrical tunability, and large internal surface areas, making them highly sought after for applications across catalysis, gas sensing, storage/separation, electrochemistry, and biomedicine. Fundamentally, MOFs are a synergistic combination of organic and inorganic building blocks. Well-understood organic linkers coordinate with metal nodes to form structures exhibiting emergent properties not found in their individual components. The inherent tunability of MOFs arises from the modularity of their construction, allowing for diverse combinations of organic linkers and inorganic nodes that can be interchanged within a given topological framework. This strategy, known as isoreticular chemistry, provides MOFs with an exceptional degree of freedom for structural engineering while conserving the underlying network architecture. In this research, we apply the bioisosteric replacement strategy (BIS), a strategy used in medicinal chemistry for changing the material composition to find new properties without altering the overall framework. Using BIS, the achiral ditopic ligand (L1) in our pore-space partitioned MOF (pacs-MOF) was substituted with a chiral counterpart, (1R,3S)-(+)-Camphoric acid (D-cam). Although D-cam has been widely employed in the construction of chiral MOFs, this study marks its first application within a pacs-MOF framework. Chiral MOFs are primarily utilized for applications such as enantioselective separation, asymmetric catalysis, and chiral

recognition. While these functions are well-established, the development of new chiral MOFs offers the potential to discover novel properties that may be useful in other applications. This research contributes by synthesizing and characterizing a specific chiral pacs-MOF, thereby laying the groundwork for future studies into its potential applications.

This work was supported by National Science Foundation, Division of Materials Research, under Award # 2105961 and by CSULB RSCA award for 3-unit teaching release time.

99. Microbial Electrolysis Cells:turning waste into green energy

Daniel Duarte Toledo, Dr. Jiajia Chen

Microbial Electrolysis Cells (MECs) offer a sustainable pathway for hydrogen production by coupling microbial metabolism with bioelectrochemical processes. In this study, aqueous effluent from hydrothermal liquefaction (HTL) of photosynthetically cultivated microalgae was employed as feedstock, linking wastewater remediation with renewable energy generation. At the anode, electrogenic bacteria oxidized organic compounds and transferred electrons via conductive nanowires, while a small applied voltage (–0.2 V) enabled proton reduction at the cathode to generate hydrogen gas.

Two MECs (A and B), inoculated with identical microbial consortia, were operated under anaerobic conditions and fed daily with algal HTL effluent. Performance was assessed through current density, Chemical Oxygen Demand (COD) removal, and Coulombic efficiency (CE). MEC A consistently outperformed MEC B, achieving maximum current densities of 50 mA compared to 27 mA, with higher COD removal and CE. The discrepancy was attributed to oxygen exposure in MEC B, which disrupted the electrogenic community and allowed methanogens to compete for substrate. This observation aligns with prior studies emphasizing the importance of strict anaerobic control in MEC systems (Logan et al., 2008; Borole, 2015).

Subsequent trials with new reactors, assembled under more rigorous anaerobic conditions and inoculated from MEC A, demonstrated improved reproducibility and performance. These findings underscore the potential of MECs to valorize algal HTL aqueous waste streams into renewable hydrogen fuel (Mehariya et al., 2021).

100. DIY– A novel 3D printing leaking-free medical device

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Gel electrophoresis is a technique used to separate deoxyribonucleic acids (DNA) and ribonucleic acids (RNA) from certain samples, by utilizing a device that has positively and negatively charged ends, allowing the negatively charged DNA and RNA to move through the gel towards the positively charged side. An ongoing challenge that researchers are still trying to overcome is the limited availability to a gel electrophoresis device, since they are high cost, and special training is needed to operate one. To investigate this, we used software to design a three-dimensional (3D) model of a gel electrophoresis device and a 3D printer to produce it. Then, the device was tested by placing the gel, specifically agarose, into its designated area along with the samples into their respective wells. Additionally, a buffer solution was added into the device, and an electric current at a specific voltage passed through the gel for a predetermined amount of time. After the experiment finishes running, we hope to find several bands at various locations in the gel, which would indicate that the sample we used contained multiple DNA fragments. Also, it should be noted that, the smaller the fragments, the farther it would move down the gel. To conclude, gel electrophoresis is a powerful tool when trying to decide the number and size of DNA fragments in a sample. In the future, we hope to investigate the underlying causes of epigenetic modifications and aid healthcare professionals in diagnosing patients who are at risk earlier.

This project was supported by the UROP/BUILD program, CSULB Mini-Grants and Summer Stipends (MGSS) Grant.

101. Pedestrian Route Choice: What Factors Influence Pedestrians to Take an Alternate Route?

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Understanding pedestrian dynamics allows scientists to study crowd management when considering safety and infrastructure design. In this project, we studied what affects pedestrians' route choice between staying on a designated route, or taking a longer, alternate route when encountering a crowded passageway. Empirical data was obtained via real life recordings collected between November 9th and November 16th, 2019, from 6:00pm to 12:00am midnight during the Eindhoven GLOW festival in the Netherlands. After isolating pedestrians travelling in the same direction, we explored crowd velocity, path density, and neighboring density during the point of decision as possible pedestrian route choice factors. Correlations between these factors and pedestrian choosing the alternate route were then analyzed using a correlation matrix and a scatterplot matrix. Following our inspection of both matrices, we concluded all the factors chosen are independent of pedestrians' choice to take an alternate route.

102. Bridge Number of Knots and Zero Forcing of Graphs

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Our research demonstrates a connection between knot theory and graph theory. Knot theory is the study of closed loops in three dimensions. Graph theory is the study of vertices and edges which denote a relation between two vertices. Historically, coloring games played on knot diagrams give information about the structure of the knot group. Coloring games on graphs have applications to optimization problems. The wirtinger coloring game on knots begins with some number of strands that are initially colored, and some that are not. An uncolored strand becomes colored using the following rule: if at a given crossing the over-strand and one of the under-strands is colored while the other under-strand is not, then the uncolored under-strand becomes colored. The wirtinger number of a knot diagram is the minimum number of initially colored strands for which there exists a sequence of wirtinger coloring moves that results in the entire knot diagram being colored. Dr. Blair's research on the wirtinger number of knots presents affirmative evidence for the Meridional Rank Conjecture (MRC). The MRC asks whether the bridge number of a knot in [Equation] is equal to the minimal number of meridians needed to generate the fundamental group of its complement. Dr. Blair and his collaborators have proven that the wirtinger number of a knot is equal to its bridge number regardless of presentation.

The zero forcing coloring game on a graph G begins with some vertices initially colored and some that are not. An uncolored vertex becomes colored using the following rule: if there exists a colored vertex v whose unique uncolored neighbor is u, then u becomes colored by v. The zero forcing number is the minimum number of initially colored vertices for which there exists a sequence of zero forcing coloring moves that results in the entire graph being colored. The zero forcing number of G is an upper bound for the maximum multiplicity of any square matrix in a family of matrices naturally associated with the graph. Dr. Ryan Blair and I present evidence to support that the wirtinger number of a knot diagram is equal to the zero forcing number for a certain naturally graph associated with the knot diagram. Such a connection would allow for application from knot theory to graph theory.

This project is supported in part by Academic Affairs and the Office of Research & Economic Development at California State University, Long Beach via the 2025 CSULB Student Summer Research Award.

103. Investigating the Role of Two Bolting-Induced Genes *bZIP34* and *bZIP61* in Leaf Senescence and Response to Climate Change Stressors

Adrian Lopez-Lopez, Destiny Holmes and Judy Brusslan

Leaf senescence (LS) is the final stage of leaf development involving nutrient redistribution from older leaves to developing leaves and tissues. Chlorophyll (Chl) degradation is also induced, resulting in a progressing yellow phenotype that starts at the leaf tip in Arabidopsis thaliana (Zou et al., 2009). LS occurs naturally due to internal signals–bolting and aging–as well as external signals like biotic and abiotic stressors (Buchanan-Wollaston et al., 2002; Balazadeh et al., 2008). Bolting is the transition from a plant's vegetative to its reproductive phase.. LS is positively associated with bolting in that early bolting leads to early LS and delayed bolting leads to delayed LS (Huang et al., 2019; Balazadeh et al., 2008). A bolting-induced gene regulatory network was predicted by Hinckley and Brusslan in 2020 showing *bZIP34*, and *bZIP61* playing a regulatory role in bolting-induced LS. During LS, reactive oxygen species, abscisic acid, and ethylene production increases (Huang et al., 2019; Gao et al., 2016; Li et al., 2013). The same molecules also play a role in response to climate change stressors such as soil salinity and drought (Peng et al., 2014; Lingqiang et al., 2000; Yoshida et al., 2002). In a meta-analysis using publicly available RNA-seq data, expression of *bZIP34* and *bZIP61* were shown to decrease in response to salt stress (Shintani et. al, 2024). Additionally, *PdbZIP61* has been shown to play a protective role in poplar trees (Lei et al 2024). We have isolated two double mutant (DM) *bzip34bzip61 Arabidopsis* lines and demonstrated they are bolting-induced genes (BIGs); however, they do not play a role in LS since bolting-induced LS is normal in bzip34bzip61 lines. Additionally, their role in response to abiotic stressors in *Arabidopsis* is unknown. Currently, we are characterizing responses to soil salinity and drought stressors in *bzip34bzip61* lines. For salt protocols, we are measuring whole plant weight, leaf 3 (L3) weight, and expression of *NIT2* and *SOS1*, markers of LS and salt-stress response, respectively. Similarly, for drought response, we are measuring whole plant and L3 weight, and *NIT2* and *DREB2A* expression, a drought-stress marker. Our results showed no significant difference in *bzip34bzip61* stress response to salt treatments when compared to WT. Drought treatment results demonstrate an increase in *DREB2A* expression in the mutant lines compared to WT, and no difference in the drought groups. We have also measured *bZIP34* and *bZIP61* expression in WT under drought stress, showing a decrease in the expression of both genes in response to the stressor. The data suggests *bZIP34* and *bZIP61* are bolting-induced genes; however, they do not play a role in regulating LS. Additionally, *bZIP34* and *bZIP61* play a role in response to drought conditions, possibly repressing drought-responsive genes under control conditions, but not to saline conditions.

104. Teachers' Knowledge and Use of Representations for Teaching Fractions and Decimals

Bethanie Ceja, Dr. Pavneet Kaur Bharaj

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Integrating multiple representations is central to developing conceptual understanding in mathematics classrooms. Although research has examined teachers' pedagogical content knowledge, less attention has been given to the specific practice of selecting and using pedagogical representations. This study evaluates the validity evidence for a new measure, the Assessing the Structure of Knowledge in Teaching Mathematics (ASKTM)-Representations, which was designed to capture teachers' knowledge of representations for teaching fractions and decimals. Using established standards for educational and psychological testing, data were collected from a random sample of elementary in-service teachers working in schools across a geographically and economically diverse Midwestern state. Additional insights were drawn from 30 think-aloud interviews in which teachers explained their reasoning as they responded to survey items. Analysis employed Delaney's (2007) qualitative coding scheme to assess the consistency of teacher reasoning. Findings highlight both the complexity of teachers' knowledge of representations and raise questions about how well current preparation programs equip teachers to effectively use drawn models in their instruction.

Funding/Acknowledgments: This material is based upon work supported by the National Science Foundation under Grant No. 1561453. Any opinions, findings, and conclusions or recommendations expressed in this material are those of the author(s) and do not necessarily reflect the views of the National Science Foundation.

105. Orchestrating Talk: How Preservice Teachers Ask Questions in Virtual Teaching Simulations

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Effective questioning anchors productive mathematical discussion, yet we know little about how elementary pre-service teachers (PSTs) enact questioning in simulated teaching. This study analyzes PSTs who, as part of a larger project led avatar-mediated discussions in a Virtual Teaching Simulator (VTS) on a Grade-4 topic: finding fractions between two given fractions. Prior to teaching, PSTs received a task packet with worked solution strategies and pedagogical guidance from their teacher educators. Sessions were recorded, transcribed, and analyzed. Using a deductive–inductive approach, we began with Boaler and Brodie's (2004) framework as a scaffold, then open-coded four transcripts (yielding 39 codes). Applying the codebook to four additional transcripts added 5 codes (total 44); coding four more transcripts produced no new codes. Related codes were organized into thematic

families. Two authors coded independently, met weekly to resolve discrepancies, and achieved strong reliability (ICC = .91). Preliminary frequencies (percent of coded segments) indicate heavy reliance on elicitation and orchestration: probing for explanation (8.87%), prompting reflection on strategy use (7.68%), affirming (5.60%), guided solution (4.76%), eliciting agreement/disagreement (3.72%), exploring mathematical meanings/relationships (3.27%), and revoicing (2.93%). In contrast, moves central to disciplinary argumentation were less common: asking students to compare strategies (2.03%), asking one group to defend another's idea (1.19%), building consensus (1.88%), and connecting student ideas—linking strategies (0.94%), restating for coherence (0.45%), connecting representations (0.15%). We argue that VTS, paired with guided analysis, yields actionable records of practice that reveal leverage points for preparation—namely, planning and rehearsal tools that cue conflict-eliciting sequences, require explicit linking across ideas, and shift talk toward multi-turn, student-driven argumentation.

This project is supported in part by National Science Foundation, Grant #2037983.

106. Toxic Threat - Bisphenol A - are frozen food safe to eat?

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Bisphenol A (BPA) is a chemical compound commonly found in plastics used for various consumer products. People are exposed to BPA regularly through food packaging, water bottles, receipts, grocery bags, dental materials, and electronics. BPA is known to disrupt hormones, damage DNA, increase the risk of insulin resistance, and impact pathways associated with cancer. This study explores a simple and reliable method to detect BPA by utilizing its reaction with a specific colorimetric reagent, which produces noticeable color changes. Tests conducted on a variety of packaged frozen foods demonstrate the method's sensitivity in detecting even trace amounts of BPA, even below levels considered harmful. In addition to its effectiveness, this colorimetric approach is simple, affordable, and highly accessible compared to conventional detection techniques. Overall, this study introduces a straightforward and efficient method to determine if packaged frozen foods are BPA-free, enhancing consumer transparency and awareness.

This project was supported by the UROP/BUILD program, CSULB Mini-Grants and Summer Stipends (MGSS) Grant and the KURE program.

107. A Characterization of Sounds from Surfers, Swimmers, and Kayakers with a Preliminary Look on Juvenile White Shark (*Carcharodon carcharias*) Response within an Aggregation Site

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Recovery of some shark populations due to conservation success and rising popularity in ocean water recreational activities has resulted in higher likelihoods of shark-human interactions, raising concerns for beach safety. However, little is known about how sharks detect and respond to human activities in coastal habitats. Juvenile white sharks (*Carcharodon carcharias*; JWS) have been observed orienting toward people from distances exceeding beyond the abilities of their near-field senses, suggesting strong directional acoustic–lateralis cues. To investigate these stimuli, we recorded swimmers, surfers, and kayakers using a particle motion–sensing hydrophone at two standardized periodicities to then characterize in the context of shark hearing. Activity recordings had particle acceleration magnitudes exceeding ambient noise levels by >20 dB at distances up to 16 m and peak energy was in the 200-800Hz range for all activities, matching our current understanding of hearing sensitivity in elasmobranchs. Acoustic indices further highlighted differences in activity sounds based on specific spectral and temporal features. These recordings were then compiled into playback experiments conducted within a natural aggregation site to assess JWS responses. Together, these findings provide novel insight into how JWS interact with people in their environment.